

Hybrid Quantum-Classical Hierarchical Actor-Critic for Sum-Rate Maximization in IRS-Assisted Grouped RSMA Systems

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Abstract

Integrated satellite-terrestrial networks (ISTNs) have emerged as a key architecture for next-generation wireless systems, offering wide-area coverage and connectivity by combining satellite and terrestrial infrastructure. However, ISTNs face fundamental challenges including severe path loss, atmospheric fading, building blockage, and the worst-user bottleneck caused by users with poor channel conditions. To address these challenges, this paper considers an IRS-assisted grouped rate-splitting multiple access (RSMA) downlink system in ISTN, where multiple intelligent reflecting surfaces (IRSs) improve link reliability for users under blockage, and grouped RSMA enables flexible interference management across heterogeneous user groups.

We propose a hybrid quantum-classical hierarchical actor-critic (HQC-HAC) framework to jointly optimize four coupled decision variables: IRS-assisted link assignment, IRS phase-shift configuration, transmit power allocation, and common-rate distribution. These variables form a sequential dependency chain — the user grouping from link assignment constrains phase-shift design, which determines effective channels for power allocation, and the resulting power levels set the rate budget for common-rate splitting. This inherent ordering motivates a hierarchical decomposition. At the top level, a quantum actor based on variational quantum circuits (VQCs) resolves the combinatorial IRS assignment bottleneck, whose search space scales as $(M+1)^K$. An autoencoder (AE) compresses the high-dimensional system state into a compact latent representation before quantum encoding to improve scalability. At the lower level, three classical sub-actors sequentially handle phase-shift, power, and common-rate decisions. Entropy regularization with an annealing schedule is applied across all actors to maintain sufficient exploration during training.

Simulation results show that the proposed HQC-HAC achieves significant sum-rate gains over baseline methods while satisfying per-user quality-of-service (QoS) constraints under imperfect channel state information (CSI) and dynamic user mobility. The quantum assignment actor attains sum-rate performance comparable to its classical

deep-network counterpart while using two orders of magnitude fewer assignment-actor parameters, and the learned policy amortizes the per-state optimization cost of model-based methods into a single offline training run.

Index Terms

Integrated satellite-terrestrial networks (ISTN), intelligent reflecting surface (IRS), rate-splitting multiple access (RSMA), hierarchical actor-critic (HAC), variational quantum circuit (VQC), hybrid quantum-classical reinforcement learning, sum-rate maximization, quality of service (QoS), resource allocation.

I. INTRODUCTION

A. Background

The proliferation of wireless services and the escalating demand for pervasive global connectivity have exposed the fundamental limitations of conventional terrestrial networks, whose coverage is inherently circumscribed by the deployment of fixed ground-based infrastructure. Satellite systems offer a compelling remedy: operating across a diverse spectrum of orbital regimes — from low Earth orbit (LEO) mega-constellations such as Starlink and OneWeb that deliver low-latency broadband, through medium Earth orbit (MEO) platforms underpinning global navigation systems, to geostationary (GEO) satellites that provide continental-scale coverage from a fixed vantage point — they maintain line-of-sight access over vast geographical expanses irrespective of terrain [1]. The recent emergence of small-form-factor CubeSats has further diversified this landscape by substantially reducing launch costs and enabling rapid constellation scaling. Integrating these orbital tiers with terrestrial infrastructure into a unified integrated satellite-terrestrial network (ISTN) has consequently attracted significant research interest as a pathway to delivering pervasive connectivity to maritime, rural, and geographically isolated environments where ground-based base stations are neither practical nor economically viable [2], [3].

Despite this coverage advantage, satellite-based communications are inherently susceptible to physical impairments that intensify with the propagation distance involved. The extended path between an orbiting satellite and a ground receiver exposes the signal to substantial free-space path loss, rain attenuation, and tropospheric scintillation [4]. In densely built environments, buildings and urban structures further compound these impairments by shadowing the direct satellite-to-user link, leaving the affected terminal without a reliable primary propagation path. Furthermore, when multiple users concurrently access the

satellite downlink, inter-user interference degrades individual link quality, and terminals operating under the most adverse channel conditions — commonly referred to as the worst-user bottleneck — impose a disproportionate constraint on the aggregate achievable rate of the entire served population [5].

The worst-user bottleneck is a well-recognized impediment in multi-user downlink design. Rate-splitting multiple access (RSMA) addresses it by introducing a structured degree of flexibility into interference management [6]. Rather than fully decoding interference, as in non-orthogonal multiple access (NOMA), or treating it entirely as noise, as in space-division multiple access (SDMA), RSMA decomposes each user message into a common component — decoded cooperatively by a designated set of receivers via successive interference cancellation (SIC) [7] — and a private component decoded independently at each terminal. This partial interference treatment effectively decouples overall system throughput from the performance of the weakest user. A grouped extension of RSMA refines this mechanism further by partitioning users into subgroups according to their channel conditions or propagation characteristics [8]: within each subgroup, the decodable common rate is bounded by the least-capable member of that group rather than by the globally worst-performing user, affording considerably finer granularity in rate allocation across the network.

Partitioning users by propagation characteristics inevitably raises the question of how to adequately serve those whose direct satellite link is geometrically obstructed. Rate-splitting alone is insufficient to compensate for a physically blocked propagation path, necessitating an alternative transmission route. Intelligent reflecting surfaces (IRSs) furnish precisely such a mechanism [9]. An IRS comprises a planar array of passive reflecting elements, each capable of independently imposing a controllable phase shift on the incident electromagnetic wave. Through judicious configuration of these phase shifts, the reflected wavefront can be coherently steered toward an intended receiver, effectively establishing a virtual line-of-sight link that circumvents the obstruction without incurring additional transmit power expenditure. In the context of grouped RSMA, each active IRS can serve as the reflective backbone of a designated user group, providing reliable alternative propagation paths for terminals that are unreachable via the direct satellite link, as illustrated in Fig. 1: Group 0 users with unobstructed satellite links receive the direct signal x_0 , while users in Group g and Group g' behind buildings are served through dedicated IRS panels carrying the reflected signals x_g and $x_{g'}$, respectively.

The joint optimization of IRS-user link assignment, phase-shift configuration, transmit power allocation,

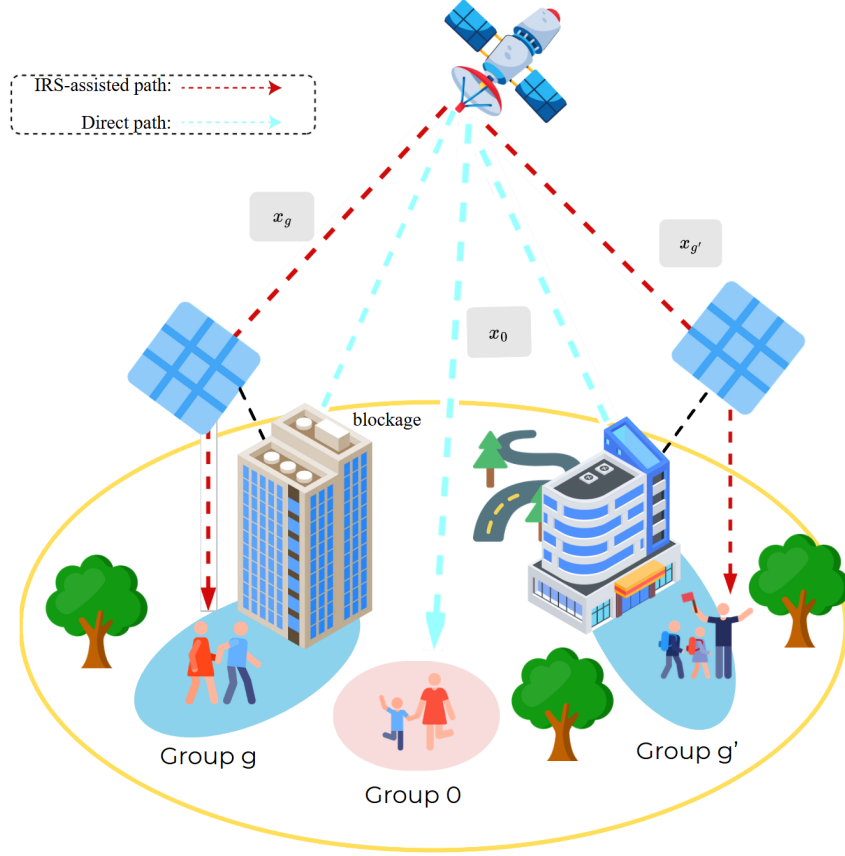


Fig. 1: Proposed IRS-assisted grouped RSMA system in an ISTN downlink. The satellite simultaneously serves $G + 1$ groups: Group 0 users with unobstructed satellite links receive the direct signal x_0 , while Group g and Group g' users blocked by buildings are served via dedicated IRS panels carrying reflected signals x_g and $x_{g'}$. Each IRS steers its reflected signal toward the assigned group through coherent phase-shift configuration Φ_m .

and common-rate distribution constitutes a mixed-integer non-convex problem that admits no tractable closed-form solution. Reinforcement learning (RL) is well-suited to this class of problems by virtue of its model-free nature, acquiring effective resource allocation policies through iterative environment interaction without requiring an explicit analytical characterization of the system [10]. A salient structural feature of the present problem is that the four optimization variables are not independent but form a strict sequential dependency: the IRS-user assignment defines the grouping structure, which in turn governs the effective channel conditions underlying the phase-shift design, which subsequently shapes the power allocation landscape, and the resulting power budget finally delimits the feasible common-rate split. This inherent ordering provides a principled motivation for a hierarchical actor-critic (HAC) decomposition [11], wherein a high-level actor commits to the discrete IRS assignment and dedicated low-level actors resolve the remaining continuous variables in sequence — an architectural paradigm demonstrated to be effective

in managing the expansive joint action spaces encountered in IRS-assisted wireless resource management [12], [13]. Entropy regularization is further incorporated across all hierarchical levels to sustain adequate policy diversity and forestall premature convergence during training [14].

Even with a hierarchical decomposition in place, the top-level IRS assignment remains the principal source of discrete complexity: as the number of deployed IRS panels and served users scales up, the space of feasible user-to-link associations expands at a rate that rapidly renders classical discrete optimization and flat RL policies intractable for realistic system dimensions. Variational quantum circuits (VQCs) offer a fundamentally different computational pathway by harnessing quantum superposition and entanglement to represent rich probability distributions over large discrete action spaces with a compact set of trainable parameters [15]. A further practical impediment arises from the high dimensionality of the wireless system state, which encompasses complex channel coefficients, blockage indicators, per-user traffic demands, and current allocation variables — a representation far too unwieldy to be encoded directly into the constrained register of an available quantum device. An autoencoder (AE) [16] bridges this dimensionality gap by learning a compact latent embedding of the system state that preserves the information most pertinent to the assignment decision, following the hybrid quantum-classical paradigm established in [17]. This AE-VQC architecture constitutes the quantum actor at the apex of the proposed HQC-HAC framework, resolving the combinatorial bottleneck at the highest level of the hierarchy while remaining fully interoperable with the classical sub-actors governing the lower-level continuous decisions. Motivated by the complementary strengths of these components, this paper proposes the hybrid quantum-classical hierarchical actor-critic (HQC-HAC) framework for sum-rate maximization in IRS-assisted grouped RSMA systems within ISTN communications.

Based on the comparison in Table I and to the best of our knowledge, no existing study jointly addresses IRS-assisted grouped RSMA, blockage-aware transmission, imperfect CSI, multi-IRS deployment, discrete phase shifts, RL-based optimization, and quantum-enhanced policy learning within a single framework. Motivated by this collective gap, this paper proposes the hybrid quantum-classical hierarchical actor-critic (HQC-HAC) framework for sum-rate maximization in IRS-assisted grouped RSMA-enabled ISTN systems.

TABLE I: Comparison of existing works on IRS/RIS-assisted communication systems

Ref.	Access	Blockage	CSI	Steer. Vec.	Prop. Path	IRS/RIS			RL	QC	
						Scale	Type	Phase			
2022 [18]	SDMA	✓	Imperf.	–	Multi	Single	Passive	Cont.	–	–	S
2022 [19]	TDMA	✓	Perf.	✓	Multi	Single	Passive	Cont.	–	–	
2024 [20]	H-NOMA	–	Perf.	✓	Single	Single	Passive	Cont.	–	–	
2025 [21]	RSMA	–	Imperf.	–	Multi	Single	Passive	Disc.	–	–	
2025 [22]	RSMA	–	Perf.	–	Single	Single	Passive	Disc.	–	–	
2025 [23]	MU-BF	–	Perf.	–	Multi	Single	Passive	Cont.	–	–	
2025 [24]	NOMA	–	Perf.	✓	Multi	Single	Passive	Cont.	✓	–	
2025 [25]	RSMA	✓	Perf.	–	Single	Multi	Passive	Cont.	–	–	S
2026 [26]	RSMA	–	Imperf.	✓	Multi	Multi	Passive	Cont.	–	–	
2026 [27]	RSMA	–	Imperf.	–	Multi	Single	Active	Cont.	–	–	S
2026 [8]	RSMA	–	Perf.	–	Single	Single	Passive	Cont.	–	–	S
Proposed	G_RSMA	✓	Imperf.	✓	Multi	Multi	Passive	Disc.	✓	✓	S

B. Contributions

Motivated by the identified gaps, this paper proposes the HQC-HAC framework for sum-rate maximization in IRS-assisted grouped RSMA systems over ISTN downlinks. The framework jointly optimizes four sequentially coupled decision variables — IRS-user link assignment, IRS phase-shift configuration, transmit power allocation, and common-rate distribution — under per-user QoS constraints, discrete phase quantization, imperfect CSI, and building blockage. The main contributions of this paper are summarized as follows:

- We formulate a sum-rate maximization problem for a multi-IRS-assisted grouped RSMA downlink system in ISTN, in which IRS-user link assignment, IRS phase-shift configuration, transmit power allocation, and common-rate distribution are jointly optimized as a sequential dependency chain under per-user QoS, discrete phase quantization, building blockage, and imperfect CSI constraints.
- We propose the HQC-HAC framework to solve the resulting mixed-integer non-convex problem by exploiting the inherent sequential dependency of the decision variables. A quantum actor based on a variational quantum circuit (VQC) resolves the combinatorial IRS-user assignment at the high level, while three classical sub-actors sequentially handle phase-shift, power, and common-rate decisions at the low level. Entropy regularization with an annealing schedule is incorporated across all hierarchical levels to sustain policy diversity and forestall premature convergence.
- We design a hybrid AE-VQC quantum actor architecture in which an autoencoder (AE) compresses the high-dimensional wireless system state — comprising channel coefficients, blockage indicators,

user demands, and allocation variables — into a compact latent representation before quantum encoding, enabling the VQC to operate within the constrained register of near-term quantum devices while preserving the information critical to the assignment decision.

C. Organization

The remainder of this paper is organized as follows. Section II reviews related works on IRS-assisted satellite communications, grouped RSMA, and hybrid quantum-classical reinforcement learning. Section III presents the system model and formulates the sum-rate maximization problem. Section IV details the proposed HQC-HAC framework, covering the hierarchical actor-critic decomposition, AE-based latent state compression, VQC-based quantum actor design, and entropy regularization schedule. Section V presents simulation results and performance comparisons against baseline methods. Section VI discusses the implications and limitations of the proposed framework. Section VII concludes the paper.

II. RELATED WORKS

A. IRS-Assisted Grouped RSMA for Sum-Rate Optimization in ISTNs

The foundational framework for IRS-assisted wireless networks was established by Wu and Zhang [28], who studied an IRS-aided multiuser MISO downlink in which a multi-antenna access point serves multiple single-antenna users via a passive reflective array. Their work demonstrated that jointly optimizing active transmit beamforming at the access point and passive phase shifts at the IRS yields substantial reductions in total transmit power relative to IRS-free baselines. By aligning the phase shifts so that the IRS-reflected signal combines coherently with the direct path at the intended receiver, the IRS effectively establishes a virtual line-of-sight link that circumvents physical obstructions and augments received signal power without additional transmit energy expenditure. Reflected signals can simultaneously be steered destructively at non-intended receivers, furnishing an additional spatial degree of freedom for interference suppression beyond what is attainable through active beamforming alone. The resulting non-convex problem was addressed via semidefinite relaxation (SDR) and alternating optimization, with analysis confirming that the required transmit power decreases on the order of N^2 as the number of reflecting elements N grows — a favorable scaling law that motivates large-element IRS deployments in practical systems.

Building upon this beamforming paradigm, Tan et al. [8] extended IRS-assisted transmission to satellite downlink communications by proposing a flexible grouped RSMA scheme for sum-rate maximization in ISTNs. In their system, the K ground users are dynamically partitioned into two disjoint groups based on real-time channel conditions and potential line-of-sight obstructions: group $\mathcal{K}_1$ users are exclusively served via the satellite-IRS-user reflected link, while group $\mathcal{K}_2$ users communicate through the direct satellite-user link. Within each group, RSMA decomposes every user's message into a common sub-message — decoded cooperatively by all group members via successive interference cancellation (SIC) using a shared codebook — and an independent private sub-message decoded solely at the intended terminal. Critically, the decodable common rate within each group is bounded by the weakest member of that subgroup rather than by the globally worst-performing user across the entire population, substantially alleviating the worst-user bottleneck inherent in conventional flat RSMA designs. The joint optimization problem encompasses satellite precoding beamforming, IRS passive phase-shift configuration, rate-splitting allocation, and binary IRS-user group selection, and was decomposed into four subproblems solved respectively via SDR with polyblock outer approximation (POA), Riemannian manifold conjugate gradient (RMCG) descent on the IRS phase manifold, alternating optimization for rate allocation, and simulated annealing for the binary group assignment, with the joint solution recovered iteratively.

Despite reporting approximately 29% sum-rate gain over the IRS-assisted RSMA baseline, the framework of [8] carries several limitations pertinent to realistic deployment. The optimization presupposes perfect and instantaneous CSI at the satellite, whereas practical satellite links are susceptible to estimation errors arising from channel aging, Doppler effects, feedback delay, and limited feedback bandwidth. Only a single IRS panel is considered, which is insufficient to extend reflected coverage across spatially dispersed user groups. Furthermore, the iterative model-based solver incurs computational complexity that scales prohibitively as the numbers of users, IRS elements, and control variables grow, and the framework provides no mechanism for adaptive policy adjustment under dynamic channel conditions or user mobility. The proposed HQC-HAC framework addresses these deficiencies by optimizing IRS-user link assignment across multiple IRS panels under imperfect CSI, building blockage, and dynamic user mobility, replacing model-based sub-problem decomposition with a hierarchical reinforcement learning approach whose variational quantum policy representation manages the combinatorial complexity of multi-IRS user assignment.

B. Entropy-Regularized Hierarchical Reinforcement Learning and Hybrid Quantum-Classical Policies

Hierarchical actor-critic (HAC) methods decompose complex decision problems into multiple levels of abstraction, with high-level actors committing to sub-goals that low-level actors subsequently fulfill [11], [29]. This temporal and functional decomposition has been demonstrated to accelerate learning in environments with large composite action spaces, including IRS-assisted wireless resource management, where jointly optimizing discrete and continuous variables over extended interaction horizons is otherwise intractable [12]. Entropy regularization is commonly incorporated into actor-critic frameworks to encourage sufficient policy diversity during training: the maximum-entropy reinforcement learning objective of the soft actor-critic algorithm [14] penalizes premature collapse to deterministic policies and has been applied extensively in continuous-action wireless optimization problems.

Hybrid quantum-classical reinforcement learning extends this paradigm by replacing or augmenting classical policy networks with variational quantum circuits (VQCs) [15]. VQCs leverage parameterized quantum state transformations to model complex nonlinear policy mappings using comparatively compact trainable representations, thereby motivating their investigation for high-dimensional combinatorial decision problems. However, directly encoding high-dimensional wireless system states into a quantum register is impractical given current hardware constraints. Nagy et al. [17] addressed this by jointly training an autoencoder (AE) with a VQC-based policy, where the AE compresses the original observation into a low-dimensional latent space suitable for quantum encoding. Motivated by these complementary contributions, this work embeds an AE-VQC quantum actor within an entropy-regularized HAC structure, applying it to the combinatorial IRS-user assignment problem in IRS-assisted grouped RSMA over ISTN downlinks.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Problem Formulation

1) *Signal Model:* We consider an IRS-assisted grouped rate-splitting multiple access (RSMA) downlink communication system in an integrated satellite-terrestrial network (ISTN). In the considered system, a satellite serves a set of ground users with the assistance of intelligent reflecting surfaces (IRSs), which are deployed on high buildings to enhance the wireless propagation environment and support users experiencing weak or blocked direct satellite-to-user links. The satellite is assumed to act as

TABLE II: Summary of Main Notations

Symbol	Description	Symbol	Description
$\mathcal{K}$	Set of ground users	$\mathcal{M}$	Set of IRSs
K	Number of ground users	M	Number of IRSs
$\mathcal{N}$	Set of IRS reflecting elements	N	Number of reflecting elements
φ_k	IRS-selection variable of user k	$\mathcal{G}$	Set of active transmission groups
$\mathcal{K}_g$	Set of users assigned to group g	P_S	Maximum transmit power of the satellite
s_k	Original message intended for user k	$s_{c,g}$	Common stream of group g
$s_{k,p,g}$	Private stream of user k in group g	Φ_m	Phase-shift matrix of IRS m
$w_{c,g}$	Scalar precoding coefficient for $s_{c,g}$	$w_{k,p,g}$	Scalar precoding coefficient for $s_{k,p,g}$
$\mathbf{w}$	Set of scalar precoding coefficients	h	Effective channel coefficient
$\theta_{m,n}$	Phase shift of the n -th IRS element	$\mathcal{Q}_\theta$	Quantized phase-angle set
B	Number of phase quantization bits	g	True channel coefficient
$\hat{g}$	Estimated channel coefficient	Δg	CSI estimation error
x	Transmitted signal at the satellite	y_k	Received signal at user k
n_0	Additive white Gaussian noise	σ^2	Noise variance
γ	SINR for decoding the corresponding stream	$R_{k,c,g}$	Common-stream decoding rate of user k
$R_{c,g}$	Decodable common rate of group g	$C_{k,g}$	Allocated common rate for user k
$R_{k,p,g}$	Achievable private rate of user k in group g	ϵ	Atmospheric attenuation coefficient
R_k	Total achievable rate of user k	R_{tot}	Achievable system sum-rate
$R_k^{\min}$	Minimum QoS requirement of user k	λ_{GAE}	GAE bias–variance parameter
γ_d	RL discount factor	ϵ_q	QoS-penalty guard constant
ϵ_c	PPO clipping range	β	Entropy regularization coefficient
λ_D	QoS-penalty weight	G_t	GAE λ -return regression target
α_{AE}	Autoencoder loss weight	$\hat{A}_t$	Generalized advantage estimate
ρ_t	PPO importance-sampling ratio	$\boldsymbol{\eta}$	VQC trainable parameters
$V(\cdot)$	State-value function (critic)	τ	SoftmaxPQC inverse temperature (head)
c_s	SPSA perturbation step	$\mathbf{a}_t$	Action $(\varphi, \Phi, \mathbf{w}, \mathbf{C})$
$\mathbf{s}_t$	State: affinity feature vector	d_{aff}	Affinity dimension $K(M+2) + 2M$
$\mathbf{z}_t$	AE latent vector $(\in \mathbb{R}^{2n_q})$	L	Number of VQC variational layers
n_q	Number of qubits	N_Q	Number of readout observables
$\hat{\mathbf{o}}_t$	VQC measurement (readout) vector	$\gamma, \mathbf{b}_{\text{LN}}$	Encoding LayerNorm learnable affine
λ_y, λ_z	Trainable encoding scales		

the central controller, which determines the transmission strategy, resource allocation, and IRS-assisted communication mode according to the observed channel conditions and user requirements.

Let $\mathcal{K} = \{1, 2, \dots, K\}$ denote the set of downlink ground users, where K is the total number of users. The satellite intends to transmit one independent message to each user. Specifically, the message intended for user $k \in \mathcal{K}$ is denoted by s_k . Therefore, the set of scheduled user messages can be written as

$$\mathbf{s} = \{s_1, s_2, \dots, s_K\}. \quad (1)$$

We consider M IRSs deployed on the rooftops of buildings to assist the satellite downlink transmission. Let $\mathcal{M} = \{1, 2, \dots, M\}$ denote the set of IRSs. Each IRS is equipped with N passive reflecting elements, which can independently adjust the phase shift of the incident signal. The IRSs are assumed to be connected to the central controller, and their reflection coefficients are configured according to the channel conditions and user association decisions. For IRS $m \in \mathcal{M}$, the reflection matrix is defined as

$$\Phi_m = \text{diag} \left(e^{j\theta_{m,1}}, e^{j\theta_{m,2}}, \dots, e^{j\theta_{m,N}} \right), \quad (2)$$

where $\theta_{m,n}$ denotes the phase shift of the n -th reflecting element at IRS m . Let B denote the number of quantization bits. Then, each IRS reflecting element can select one phase shift from 2^B discrete phase angles. The quantized phase-angle set is defined as

$$\mathcal{Q}_\theta = \left\{ \theta^{(0)}, \theta^{(1)}, \dots, \theta^{(2^B-1)} \right\}, \quad (3)$$

where the q -th discrete phase angle is given by

$$\theta^{(q)} = \frac{2\pi q}{2^B}, \quad q = 0, 1, \dots, 2^B - 1. \quad (4)$$

To determine the transmission mode of each user, we define an IRS-selection variable for every user $k \in \mathcal{K}$. Since the considered system includes M IRSs, each user can either be served through the direct satellite-to-user link or through one of the IRS-assisted reflected links. Therefore, the IRS-selection variable is defined as

$$\varphi_k \in \{0, 1, \dots, M\}, \quad \forall k \in \mathcal{K}, \quad (5)$$

where

$$\varphi_k = \begin{cases} 0, & \text{if user } k \text{ is served through the direct satellite-to-user link,} \\ m, & \text{if user } k \text{ is served through the satellite-IRS } m\text{-user link.} \end{cases} \quad (6)$$

Following the RSMA principle, each user message s_k of user $k \in \mathcal{K}_g$ is split into a common sub-message $s_{c,g}$ and a private sub-message $s_{k,p,g}$. However, not all IRSs are necessarily selected by the agent at each decision step. Therefore, the number of active groups is denoted by G , where $G \leq M + 1$. We define the set of scheduled user messages after splitting as

$$\tilde{\mathbf{s}} = [s_{c,1}, s_{c,2}, \dots, s_{c,G}, \{s_{k,p,g}\}_{g \in \mathcal{G}, k \in \mathcal{K}_g}].$$

Due to the long propagation distance between the satellite and ground nodes, the satellite-to-user (SU) and satellite-to-IRS (SR) links are dominated by large-scale path loss and rain attenuation following the ITU-R model [30]. The IRS-to-user (RU) link additionally exhibits small-scale Rayleigh fading and distance-dependent path loss. The true channel coefficients of the three links are modeled as

$$\begin{aligned} g_k^{SU} &= \frac{c\sqrt{G_S G_U \epsilon_k^{SU}}}{4\pi f_{SU} d_k^{SU}}, & k \in \mathcal{K}, \\ g_m^{SR} &= \frac{c\sqrt{G_S \epsilon_m^{SR}}}{4\pi f_{SR} d_m^{SR}}, & m \in \mathcal{M}, \\ g_{m,k}^{RU} &= \sqrt{G_U \cdot \Omega_{sf} \cdot \left(\frac{d_0}{d_{m,k}^{RU}}\right)^\alpha \cdot \epsilon_{m,k}^{RU} \tilde{g}_{m,k}^{sf}}, & m \in \mathcal{M}, k \in \mathcal{K}, \end{aligned} \quad (7)$$

where c is the speed of light; G_S and G_U are the satellite transmit and user receive antenna gains; f_{SU} and f_{SR} are the corresponding carrier frequencies; d_k^{SU} , d_m^{SR} , and $d_{m,k}^{RU}$ denote the satellite-to-user, satellite-to-IRS, and IRS-to-user link distances; ϵ_k^{SU} , ϵ_m^{SR} , and $\epsilon_{m,k}^{RU}$ are the rain-attenuation coefficients modeled as lognormal random variables per the ITU-R standard; Ω_{sf} is the average shadow-fading power gain of the IRS-to-user link; α is the path-loss exponent; d_0 is the reference distance; and $\tilde{g}_{m,k}^{sf} \sim \mathcal{CN}(0, 1)$ represents the small-scale Rayleigh fading component. Each channel coefficient is complex, with a random propagation phase uniformly distributed on $[0, 2\pi)$, which is tracked via Doppler compensation at the satellite.

Since the satellite acquires only an imperfect estimate of each channel coefficient, the estimated channel $\hat{g}$ and the true channel g are related by a multiplicative error model:

$$\hat{g} = g + \Delta g, \quad \Delta g = \kappa |g| \varepsilon, \quad \varepsilon \sim \mathcal{CN}(0, 1), \quad (8)$$

where $\kappa \geq 0$ is the relative error coefficient applied uniformly to all three links. This yields the estimated channels $\hat{g}_k^{SU}$, $\hat{g}_m^{SR}$, and $\hat{g}_{m,k}^{RU}$, which are used for all precoding and resource allocation design. The policy observes only the estimated channel magnitudes $|\hat{g}|$ and designs the assignment and precoding on them, whereas the environment scores the achieved rate on the true channels g . Imperfect CSI therefore manifests

as a design-versus-reality gap: the $\hat{g}$ -optimal decisions are evaluated against the physical channel g .

Since a single-antenna satellite is considered, spatial beamforming is not performed. Instead, each RSMA stream is assigned a scalar precoding coefficient. For each active group $g \in \mathcal{G}$, let $w_{c,g} \in \mathbb{C}$ denote the scalar precoding coefficient for the group-specific common stream $s_{c,g}$. Similarly, let $w_{k,p,g} \in \mathbb{C}$ denote the scalar precoding coefficient for the private stream $s_{k,p,g}$ of user $k \in \mathcal{K}_g$. Based on the grouped RSMA transmission structure, the transmitted signal at the satellite is expressed as

$$x = \sum_{g \in \mathcal{G}} w_{c,g} s_{c,g} + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} w_{k,p,g} s_{k,p,g}, \quad (9)$$

where $s_{c,g}$ is the common stream intended for all users in group $\mathcal{K}_g$, and $s_{k,p,g}$ is the private stream intended only for user k in group g . The scalar precoding coefficients are constrained by the maximum transmit power of the satellite, i.e., $\sum_{g \in \mathcal{G}} |w_{c,g}|^2 + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} |w_{k,p,g}|^2 \leq P_S$, where P_S denotes the maximum transmit power of the satellite.

Fig. 2 illustrates the RSMA Encoder structure for a representative group $g \in \mathcal{G}$, which consists of three sequential stages. In the *Rate Splitting* stage, each user message $s_{k,g}$ is decomposed into a common sub-message $s_{k,c,g}$ and a private sub-message $s_{k,p,g}$. The common sub-messages of all $|\mathcal{K}_g|$ users are jointly encoded into a single group codeword $s_{c,g}$, while each private sub-message remains exclusive to its intended user. In the *Scalar Precoding* stage, the common codeword $s_{c,g}$ is scaled by the shared coefficient $w_{c,g}$, and each private stream $s_{k,p,g}$ is independently scaled by the user-specific coefficient $w_{k,p,g}$. Finally, all precoded streams are superimposed in the *Precoded Streams* stage to form the group signal x_g . The same encoder structure is applied independently to every active group $g \in \mathcal{G}$, and the total transmitted signal aggregates all group contributions: $x = \sum_{g \in \mathcal{G}} x_g$.

After receiving y_k , each user first decodes its group-specific common stream and then decodes its private stream by applying SIC. For user $k \in \mathcal{K}_g$, the SINRs for decoding the common stream $s_{c,g}$ and the private stream $s_{k,p,g}$ are respectively given by

$$\begin{aligned} \gamma_{k,c,g} &= \frac{|h_k w_{c,g}|^2}{\sum_{\substack{g' \in \mathcal{G} \\ g' \neq g}} |h_k w_{c,g'}|^2 + \sum_{g' \in \mathcal{G}} \sum_{j \in \mathcal{K}_{g'}} |h_k w_{j,p,g'}|^2 + \sigma^2}, \\ \gamma_{k,p,g} &= \frac{|h_k w_{k,p,g}|^2}{\sum_{g' \in \mathcal{G}} \sum_{\substack{j \in \mathcal{K}_{g'} \\ j \neq k}} |h_k w_{j,p,g'}|^2 + \sum_{\substack{g' \in \mathcal{G} \\ g' \neq g}} |h_k w_{c,g'}|^2 + \sigma^2}. \end{aligned} \quad (10)$$

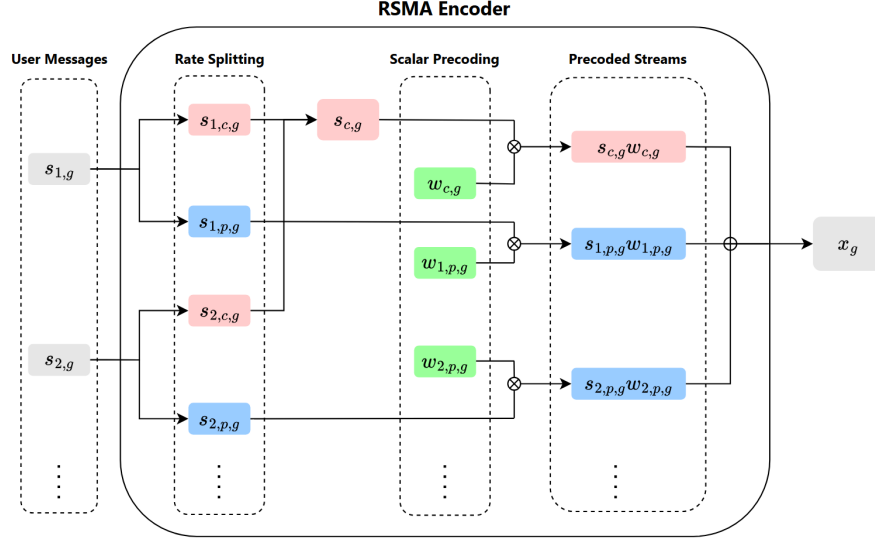


Fig. 2: RSMA Encoder structure for a representative group $g \in \mathcal{G}$, comprising three stages: *Rate Splitting* — each user message $s_{k,g}$ is decomposed into a common sub-message $s_{k,c,g}$ (jointly encoded into the shared codeword $s_{c,g}$) and a private sub-message $s_{k,p,g}$; *Scalar Precoding* — $s_{c,g}$ and each $s_{k,p,g}$ are independently scaled by $w_{c,g}$ and $w_{k,p,g}$, respectively; *Precoded Streams* — all weighted streams are superimposed to form the group signal x_g . The overall transmitted signal is $x = \sum_{g \in \mathcal{G}} x_g$.

Since the common stream $s_{c,g}$ is intended for all users in group $\mathcal{K}_g$, it must be successfully decoded by every user in that group. Therefore, the achievable common rate of group g is limited by the user with the lowest common-stream decoding rate. For user $k \in \mathcal{K}_g$, the common-stream decoding rate and the resulting group common rate are given by

$$R_{k,c,g} = \log_2(1 + \gamma_{k,c,g}), \quad R_{c,g} = \min_{k \in \mathcal{K}_g} R_{k,c,g}. \quad (11)$$

Let $C_{k,g}$ denote the portion of the group common rate allocated to user $k \in \mathcal{K}_g$. The common-rate allocation must satisfy

$$\sum_{k \in \mathcal{K}_g} C_{k,g} = R_{c,g}, \quad C_{k,g} \geq 0. \quad (12)$$

After decoding and removing the common stream, user k decodes its private stream, achieving rate $R_{k,p,g} = \log_2(1 + \gamma_{k,p,g})$. The total achievable rate of user k is thus $R_k = R_{k,p,g} + C_{k,g}$, and the system sum-rate is

$$R_{\text{tot}} = \sum_{k \in \mathcal{K}} R_k = \sum_{g \in \mathcal{G}} R_{c,g} + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} R_{k,p,g}. \quad (13)$$

2) *Optimization Problem:* Based on the above system model, the objective of this work is to maximize the achievable system sum-rate by jointly optimizing the IRS-selection variable, scalar precoding coefficients, IRS phase-shift matrices, and common-rate allocation. Specifically, the optimization variables are given by $\varphi = \{\varphi_k\}_{k \in \mathcal{K}}$, $\mathbf{w} = \{w_{c,g}, w_{k,p,g}\}_{g \in \mathcal{G}, k \in \mathcal{K}_g}$, $\Phi = \{\Phi_m\}_{m \in \mathcal{M}}$, and $\mathbf{C} = \{C_{k,g}\}_{k \in \mathcal{K}_g, g \in \mathcal{G}}$. The sum-rate maximization problem is formulated as

$$(P0): \quad \max_{\varphi, \mathbf{w}, \Phi, \mathbf{C}} \quad R_{\text{tot}} \quad (14a)$$

$$\text{s.t.} \quad \varphi_k \in \{0, 1, \dots, M\}, \quad \forall k \in \mathcal{K}, \quad (14b)$$

$$\sum_{g \in \mathcal{G}} |w_{c,g}|^2 + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} |w_{k,p,g}|^2 \leq P_S, \quad (14c)$$

$$\theta_{m,n} \in \mathcal{Q}_\theta, \quad \forall m \in \mathcal{M}, \forall n \in \mathcal{N}, \quad (14d)$$

$$\Phi_m = \text{diag}(e^{j\theta_{m,1}}, e^{j\theta_{m,2}}, \dots, e^{j\theta_{m,N}}), \quad \forall m \in \mathcal{M}, \quad (14e)$$

$$\sum_{k \in \mathcal{K}_g} C_{k,g} = R_{c,g}, \quad \forall g \in \mathcal{G}, \quad (14f)$$

$$C_{k,g} \geq 0, \quad \forall k \in \mathcal{K}_g, \forall g \in \mathcal{G}, \quad (14g)$$

$$R_k \geq R_k^{\min}, \quad \forall k \in \mathcal{K}. \quad (14h)$$

In the above problem (P0), (14b) limits the values of the IRS-selection variables, which determine whether each user is served through the direct satellite-to-user link or one of the available IRS-assisted links. (14c) guarantees that the total transmit power of the satellite does not exceed the maximum power budget. (14d) restricts the phase shift of each IRS reflecting element to the discrete quantized phase-angle set, while constraint (14e) defines the corresponding passive IRS phase-shift matrix. Constraint (14f) requires that the allocated common rates fully exhaust the group common rate, i.e., the entire decodable common rate of each group is distributed among its members with no waste. Constraint (14g) ensures non-negativity of the allocated common rates, and constraint (14h) enforces the per-user minimum QoS requirement.

IV. PROPOSED HQC-HAC FRAMEWORK

A. Overall Framework

Problem (P0) is a mixed-integer non-convex program. The IRS-selection variables φ are discrete with a combinatorial feasible set of size $(M+1)^K$; the per-element phase shifts $\theta_{m,n}$ are restricted to the quantized set $\mathcal{Q}_\theta$; and these discrete choices are tightly coupled with the continuous precoding powers $\mathbf{w}$ and common-rate shares $\mathbf{C}$ through the SINR expressions. The objective R_{tot} is non-concave, since each user's rate depends on the ratio of its effective channel gain to the aggregate cross-stream interference, and the per-user QoS constraints (14h) couple all four variable blocks simultaneously. Consequently, (P0) is generally NP-hard [31] and admits no closed-form solution. Conventional model-based methods — semidefinite relaxation [32], alternating optimization, manifold optimization, or branch-and-bound — scale poorly with K , M , and N , must be re-solved from scratch for every channel realization, and presuppose accurate instantaneous CSI that is unavailable under the imperfect-CSI and user-mobility conditions considered here. These limitations motivate a model-free reinforcement learning (RL) formulation: rather than re-solving (P0) at every coherence interval, we cast the sequential resource-allocation task as a Markov decision process and learn a policy that maps the observed system state directly to the joint action, amortizing the optimization cost into a trained network that adapts online to channel dynamics and estimation error.

The four decision variables of (P0) exhibit a strict sequential dependency: the assignment φ determines which IRS panels are active and thereby fixes the number of active groups G and the dimension of the common-power vector $\mathbf{w}_c$; the active set governs the effective channels that underpin the phase-shift design Φ ; the resulting channel gains shape the power-allocation landscape $(\mathbf{w}_p, \mathbf{w}_c)$; and the allocated powers finally delimit the feasible common-rate split $\mathbf{C}$. A flat RL agent acting over the entire joint, and mixed discrete–continuous, action space would face an action dimension that explodes with K and M and would have to learn this coupling implicitly. We instead adopt a hierarchical actor–critic (HAC) decomposition that mirrors the natural ordering of the variables: a high-level actor commits to the discrete assignment, and three low-level actors resolve the continuous variables in sequence, each conditioned on the decisions made upstream.

Even after this decomposition, the high-level assignment $\varphi \in \{0, 1, \dots, M\}^K$ remains the principal source of difficulty: its search space of size $(M+1)^K$ is non-differentiable and grows exponentially in

both M and K , so a classical policy head would require a rapidly expanding parameter budget to represent a sufficiently expressive distribution over candidate assignments. We therefore place a variational quantum circuit (VQC) at the top of the hierarchy. By preparing an n_q -qubit state that spans a 2^{n_q} -dimensional Hilbert space, the VQC represents rich, correlated distributions over the exponentially large assignment space using only $\mathcal{O}(Ln_q)$ trainable parameters. Its entangling layers — a nearest-neighbor CZ chain augmented with cross-block bridges coupling the user-qubit and IRS-qubit registers — further inject an inductive bias matched to the user–IRS coupling that the assignment must capture. Once φ is fixed, the remaining sub-problems are smooth, continuous, and low-dimensional, and are handled efficiently by classical multi-layer perceptrons (MLPs); reserving the quantum circuit for the combinatorial bottleneck is precisely what renders the framework *hybrid* quantum–classical.

A direct quantum encoding of the raw state is, however, infeasible. The raw environment observation has a dimension that scales with the number of reflecting elements N and far exceeds the register size of near-term quantum hardware; moreover, much of it — such as the full per-element phase configuration — is only weakly informative for the assignment decision. Encoding it qubit-by-qubit would thus demand an impractically large register. We bridge this gap with an autoencoder (AE) that compresses the state into a compact latent vector $\mathbf{z}_t \in \mathbb{R}^{2n_q}$, exactly matching the encoding capacity of the n_q -qubit circuit while preserving the information most relevant to link assignment. A reconstruction loss regularizes this latent space. The AE is given a brief reconstruction-only warm-up so that its latent is informative from the outset. Thereafter, it is trained jointly and end-to-end with the policy, rather than being frozen as an inference-only feature extractor. This ensures the representation remains task-aware and co-adapts with the quantum actor throughout learning.

Fig. 3 illustrates the full architecture. At each decision step t , the raw channel observation is condensed into a compact *affinity feature vector* $\mathbf{s}_t \in \mathbb{R}^{d_{\text{aff}}}$ (Eq. (15)) that encodes the end-to-end user–IRS path qualities, the per-user demands, and per-IRS summary statistics — the quantities that determine the assignment. A flat real/imaginary channel vector is a poor policy input: its dimension grows with the number of reflecting elements N and it does not expose the user–IRS spatial relationships directly, whereas $\mathbf{s}_t$ makes them explicit while discarding the N -dependent bulk. The affinity $\mathbf{s}_t$ is the state on which both the quantum actor (through the AE and VQC) and the shared critic operate; the autoencoder further compresses it to the latent $\mathbf{z}_t \in \mathbb{R}^{2n_q}$ that fits the near-term qubit register.

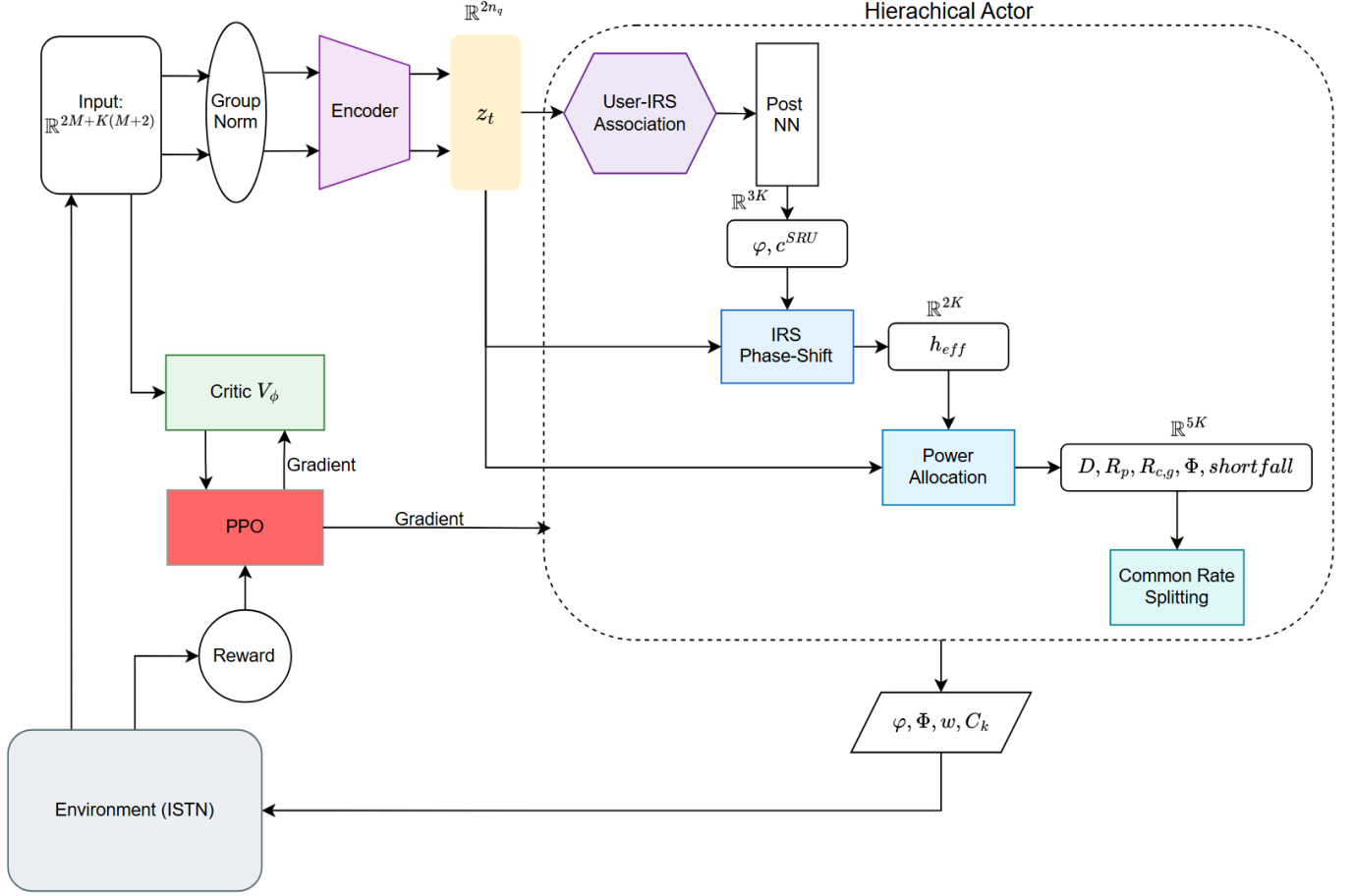


Fig. 3: Overall HQC-HAC hierarchical actor-critic framework. The quantum actor (top level) resolves the discrete IRS-user assignment φ via a VQC, while three sequential classical sub-actors — Phase MLP, Power MLP, and Common-Rate MLP — optimize the remaining continuous variables conditioned on the chosen assignment. A shared classical critic provides state-value estimates for computing advantages across all actors.

Following the data flow in Fig. 3, the latent z_t produced by the AE encoder is mapped by the VQC to a measurement vector, which the quantum actor’s output head converts into a categorical distribution over candidate assignments; sampling this distribution yields φ . Conditioned on φ , the three low-level sub-actors act in sequence: the Phase MLP produces the quantized phase indices θ_{idx} (and hence Φ), the Power MLP allocates the private and common powers (w_p, w_c) , and the Common-Rate MLP distributes the group common rates C among users. The resulting joint action (φ, Φ, w, C) is applied to the environment, which returns the next state and a reward equal to the system sum-rate minus a soft QoS-violation penalty. A single shared critic evaluates the same affinity state s_t (Eq. (15)) that the quantum actor consumes, and supplies the advantage signal used to update all four actors.

The framework is trained as an actor–critic system optimized by proximal policy optimization (PPO) [33].

The shared state-value critic $V(s_t; \psi)$ provides a low-variance, action-independent baseline for advantage estimation, which is essential here for two reasons: the quantum policy gradients, estimated by simultaneous perturbation stochastic approximation (SPSA) [34], are inherently noisy, and the reward is itself stochastic under per-step noise and imperfect CSI; subtracting a state-only baseline reduces this variance without introducing bias and avoids the self-cancellation that a state-action baseline would incur. PPO is adopted in preference to vanilla policy gradient because its clipped surrogate objective bounds the magnitude of each update, yielding trust-region-like stability that is critical in a hierarchical architecture: a large step in the high-level assignment would shift the input distribution seen by every downstream actor, and, combined with noisy quantum gradients, could readily destabilize training. The importance-sampling ratio underlying PPO additionally permits several optimization epochs over each batch of collected rollouts, improving sample efficiency — a valuable property given the cost of quantum-circuit evaluations. Entropy regularization with an annealing schedule is applied across all actors to sustain exploration over the combinatorial assignment space and to prevent premature convergence, while the per-user QoS requirement (14h) is relaxed into a soft quadratic penalty added to the reward so that demand shortfalls are penalized proportionally without blocking gradient flow. The critic is updated by regression toward a generalized-advantage-estimation (GAE) [35] λ -return target with PopArt value normalization [36], and detailed descriptions of all components — including the AE, the VQC, and the training procedure — are provided in Sections IV-B–IV-D.

B. Autoencoder-Based Latent State Representation

Directly encoding the raw environment observation into a quantum register is impractical: its dimension $d_s = M(3 + 2K + 2N) + 3K$ grows with the number of IRS elements N , which is unrelated to the IRS-user assignment decision and introduces irrelevant information into the quantum encoding. Instead, the quantum actor operates on the compact *affinity feature vector* $s_t \in \mathbb{R}^{d_{aff}}$ of Section IV, which retains only the information pertinent to link assignment, where $d_{aff} = K(M + 2) + 2M$. The affinity vector is structured as

$$s_t = \left[\underbrace{a_{k,m}}_{\substack{K \times M \\ \text{IRS-user} \\ \text{affinity}}}, \underbrace{D_k}_K, \underbrace{|\hat{g}_k^{SU}|}_K, \underbrace{q_m, p_m}_{2M} \right], \quad (15)$$

where $a_{k,m} = |\hat{g}_m^{SR}| \cdot |\hat{g}_{m,k}^{RU}|$ is the end-to-end IRS path quality for user k via IRS m , D_k is the per-user QoS demand, $|\hat{g}_k^{SU}|$ is the direct satellite-to-user link quality, $q_m = \frac{1}{K} \sum_{k=1}^K a_{k,m}$ is the mean affinity of IRS m across all users, and $p_m = |\hat{g}_m^{SR}|^2$ is the received satellite-to-IRS power. Prior to encoding, a group-wise layer normalization is applied to the affinity, direct-link, and IRS feature blocks, while the demand block D_k is kept unnormalized to preserve the absolute rate-scale information it carries.

A dual-branch encoder then compresses $\mathbf{s}_t$ into the latent vector $\mathbf{z}_t \in \mathbb{R}^{2n_q}$, where n_q is the number of qubits. As illustrated in Fig. 4, the encoder consists of two parallel branches that process IRS-level and user-level information separately before concatenation. The *IRS branch* processes the IRS summary statistics $[q_m, p_m]_{m=1}^M \in \mathbb{R}^{2M}$ through a two-layer MLP,

$$\mathbf{z}_{IRS} = \mathbf{W}_2^{IRS} \sigma(\mathbf{W}_1^{IRS} [q_1, p_1, \dots, q_M, p_M]^\top + \mathbf{b}_1^{IRS}) + \mathbf{b}_2^{IRS} \in \mathbb{R}^{2M}, \quad (16)$$

where $\mathbf{W}_1^{IRS} \in \mathbb{R}^{16 \times 2M}$, $\mathbf{W}_2^{IRS} \in \mathbb{R}^{2M \times 16}$, and $\sigma(\cdot)$ denotes the ReLU activation. The *user branch* applies a shared micro-encoder ϕ independently to each user k ,

$$\mathbf{z}_k = \mathbf{W}_2^U \sigma(\mathbf{W}_1^U [a_{k,1}, \dots, a_{k,M}, D_k, |\hat{g}_k^{SU}|]^\top + \mathbf{b}_1^U) + \mathbf{b}_2^U \in \mathbb{R}^2, \quad \forall k \in \mathcal{K}, \quad (17)$$

where $\mathbf{W}_1^U \in \mathbb{R}^{8 \times (M+2)}$ and $\mathbf{W}_2^U \in \mathbb{R}^{2 \times 8}$ are shared across all K users. The K per-user embeddings are stacked into $\mathbf{z}_{stack} = [\mathbf{z}_1^\top, \dots, \mathbf{z}_K^\top]^\top \in \mathbb{R}^{2K}$ and projected to the user-block latent subspace by a linear map $\mathbf{P} \in \mathbb{R}^{2(n_q-M) \times 2K}$, giving $\mathbf{z}_{user} = \mathbf{P} \mathbf{z}_{stack} \in \mathbb{R}^{2(n_q-M)}$. The final latent vector concatenates the two branches, $\mathbf{z}_t = [\mathbf{z}_{user} \parallel \mathbf{z}_{IRS}] \in \mathbb{R}^{2n_q}$, so that the first $2(n_q - M)$ dimensions carry user-assignment information and the last $2M$ dimensions carry IRS-quality information, matching the qubit partition of the VQC.

A symmetric decoder mirrors the encoder depth and reconstructs $\hat{\mathbf{s}}_t \in \mathbb{R}^{d_{aff}}$ from $\mathbf{z}_t$, producing the reconstruction loss

$$\mathcal{L}_{AE} = \frac{1}{2} \|\hat{\mathbf{s}}_t - \mathbf{s}_{\text{norm},t}\|^2, \quad (18)$$

where $\mathbf{s}_{\text{norm},t}$ is the group-normalized input. The decoder is used only during training and is discarded at inference time, where solely the encoder and the latent $\mathbf{z}_t$ are forwarded to the VQC.

The autoencoder is trained jointly and end-to-end with the remainder of the framework: the reconstruction loss in (18) is added to the policy objective as an auxiliary regularizer with weight α_{AE} , so that the

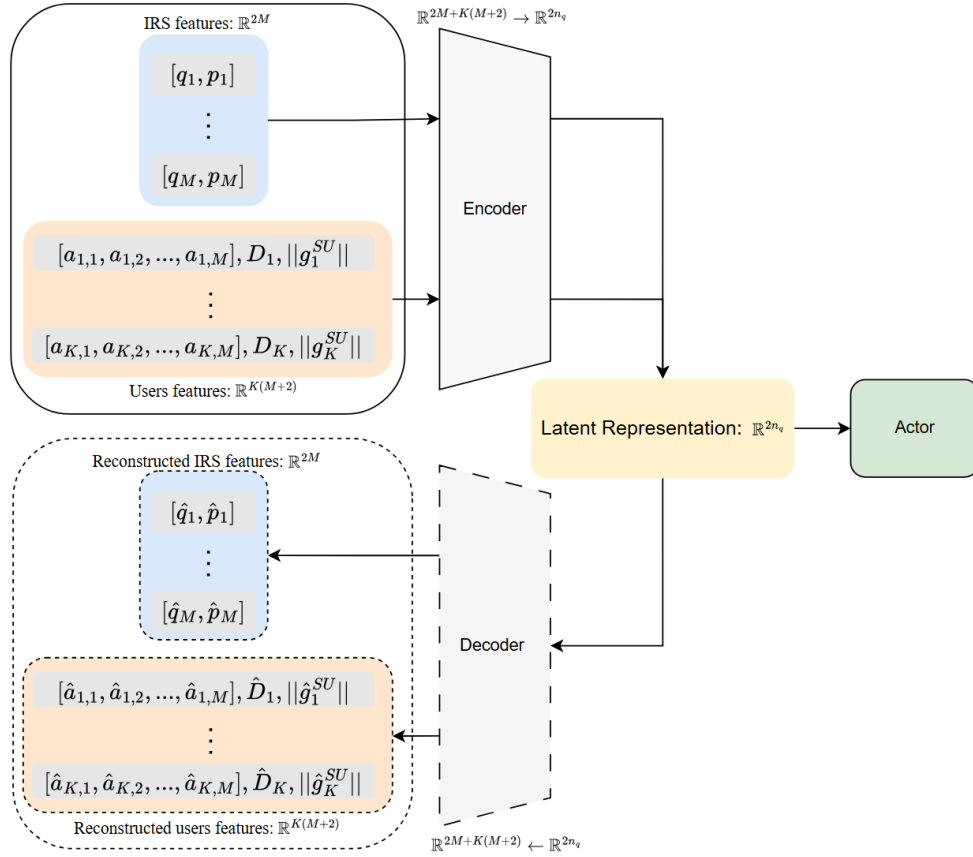


Fig. 4: Dual-branch autoencoder architecture for latent state compression. The encoder (top path) processes IRS summary features $[q_m, p_m]$ and per-user affinity features $[a_{k,m}, D_k, |\hat{g}_k^{SU}|]$ through separate branches before concatenating into the latent representation $\mathbf{z}_t \in \mathbb{R}^{2n_q}$, which is forwarded to the quantum actor. The decoder (bottom path, active during training only) reconstructs the input feature vector $\hat{\mathbf{s}}_t$ from $\mathbf{z}_t$ and computes the reconstruction loss $\mathcal{L}_{AE} = \frac{1}{2} \|\hat{\mathbf{s}}_t - \mathbf{s}_t\|^2$ to regularize the latent space.

encoder receives gradients from both $\mathcal{L}_{AE}$ and the reinforcement-learning loss back-propagated through the quantum actor. This contrasts with the common practice of training an autoencoder offline and then freezing it as a fixed feature extractor for inference: here the encoder is never frozen, and continues to adapt throughout the reinforcement-learning phase. End-to-end training is preferred for two reasons. First, reconstruction fidelity is not the true objective: a latent optimized solely to reconstruct its input may retain features irrelevant to link assignment while discarding decision-critical ones, whereas end-to-end training lets the policy gradient shape $\mathbf{z}_t$ to be *task-aware*, emphasizing precisely the information that improves the assignment and the resulting sum-rate. Second, the distribution of visited states is non-stationary: as the policy improves, the states it encounters drift, so a frozen latent would become mismatched to the current policy, whereas a jointly trained encoder co-adapts with the evolving policy and the downstream VQC and sub-actors. The reconstruction term is retained throughout because the reinforcement-learning gradient

alone can collapse the latent space to a degenerate representation; $\mathcal{L}_{AE}$ thus acts as a structural regularizer that keeps $\mathbf{z}_t$ informative, with α_{AE} trading off representational fidelity against task performance.

To prevent the encoder from lagging behind the faster-adapting policy networks at the start of training, a short reconstruction-only *warm-up* is performed on states collected under a random policy before joint optimization begins. Without this warm-up, a randomly initialized encoder would emit an uninformative latent in the early episodes, effectively feeding noise into the VQC and retarding the initial learning of all downstream actors. The warm-up mitigates this by supplying a sensible starting representation, placing the AE on a comparable footing with the other components. We emphasize that this step is purely an initialization and *not* an offline pre-training of a frozen feature extractor: once joint optimization begins, the encoder is not frozen but continues to be trained together with the policy, so the latent evolves from this reconstruction-faithful starting point into a task-aware representation.

C. Variational Quantum Circuit

The latent vector $\mathbf{z}_t \in \mathbb{R}^{2n_q}$ from the dual-branch encoder is processed by a variational quantum circuit (VQC) acting on an n_q -qubit register, which maps it to a compact feature vector $\hat{\mathbf{o}}_t$. The dimension $2n_q$ supplies one rotation angle per qubit along each of two encoding axes (defined below). By exploiting quantum superposition and entanglement, the VQC represents correlations among the user–IRS affinities with a compact trainable-parameter budget (Section V-B2).

1) *Data Encoding*: Before encoding, the latent vector is layer-normalized [37] to stabilize the input range:

$$\tilde{\mathbf{z}}_t = \gamma \odot \frac{\mathbf{z}_t - \mu_z}{\sigma_z + \epsilon} + \mathbf{b}_{\text{LN}}, \quad (19)$$

where μ_z and σ_z are the mean and standard deviation of $\mathbf{z}_t$, and $\gamma, \mathbf{b}_{\text{LN}} \in \mathbb{R}^{2n_q}$ are a learnable affine trained jointly with the quantum parameters, restoring the DC component that a plain zero-mean LayerNorm would annihilate. The normalized vector is then mapped to per-qubit rotation angles via trainable encoding scales $\lambda_y, \lambda_z \in \mathbb{R}^{n_q}$:

$$\alpha_i = \pi \tanh(\lambda_i^y \cdot \tilde{z}_{2i}), \quad \delta_i = \pi \tanh(\lambda_i^z \cdot \tilde{z}_{2i+1}), \quad i = 0, \dots, n_q - 1. \quad (20)$$

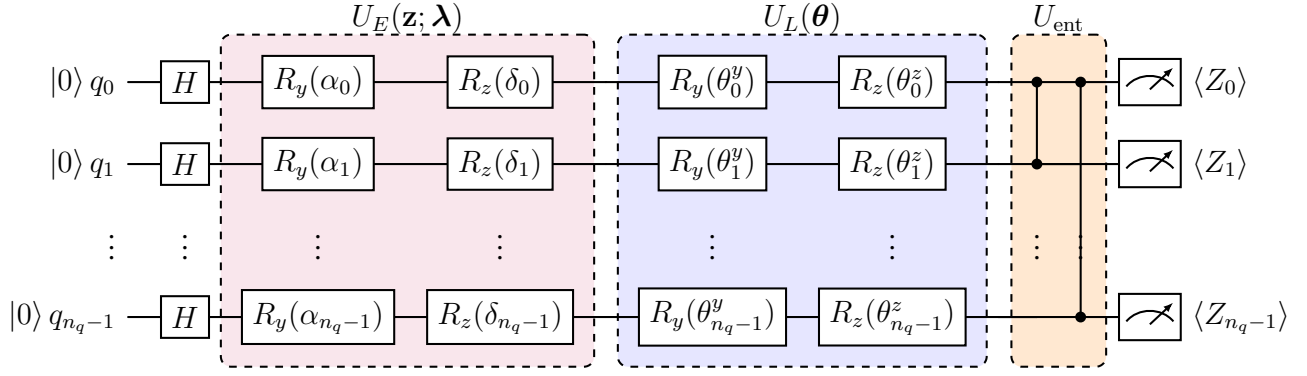


Fig. 5: Variational quantum circuit (VQC) of the quantum actor, shown for representative qubits $q_0, q_1, \dots, q_{n_q-1}$. After $H^{\otimes n_q}$, each of the L layers applies the encoding block U_E with $\alpha_i = \pi \tanh(\lambda_i^y \tilde{z}_{2i})$ and $\delta_i = \pi \tanh(\lambda_i^z \tilde{z}_{2i+1})$, the variational block $U_L(\theta)$, and the entangling block U_{ent} (nearest-neighbor CZ chain plus a cross-block bridge linking the user- and IRS-qubit registers). The encoding block is re-applied at each layer (data re-uploading). The Pauli expectations $\langle Z_i \rangle$ and $\langle Z_i Z_j \rangle$ form the measurement vector $\hat{\mathbf{o}}_t$.

The $\tanh$ squashes each angle to $(-\pi, \pi)$, preventing gradient vanishing from over-rotation. The trainable scales λ allow the circuit to selectively amplify or suppress each latent dimension's influence on the quantum state, and are optimized jointly with the variational parameters via SPSA (detailed in Section IV-D).

2) *Circuit Architecture*: As illustrated in Fig. 5, the register is first placed in a uniform superposition by a layer of Hadamard (H) gates, after which L layers are applied, each composed of three blocks built only from single- and two-qubit gates. The *encoding block* U_E applies a pair of single-qubit rotations $R_y(\alpha_i)$ and $R_z(\delta_i)$ to every qubit; two axes are used because an $R_y R_z$ pair can steer a qubit to an arbitrary point on the Bloch sphere, so the $2n_q$ latent components are loaded into the quantum state without loss. The *variational block* U_L applies trainable rotations $R_y(\theta_i^y)$ and $R_z(\theta_i^z)$ on each qubit; these are the free parameters that the policy gradient tunes and, together with the entangler, form a universal layered ansatz. The *entangling block* U_{ent} uses controlled- Z (CZ) gates: a nearest-neighbor CZ chain builds correlations beyond product states, augmented by M cross-block CZ bridges. These bridges are the key problem-specific design choice — since the latent is partitioned so that its first $2(n_q - M)$ components carry user information and its last $2M$ carry IRS information, the bridges let the circuit model the user–IRS coupling that governs the assignment, which a within-block chain alone cannot represent. Finally, U_E is re-applied at the start of every layer (*data re-uploading* [38]); re-injecting $\mathbf{z}_t$ at each depth enlarges the function class the circuit can express and keeps the input information from being washed out by successive entangling layers. The circuit therefore relies solely on the gates H , R_y , R_z , and CZ, which keeps it hardware-efficient.

3) *Measurement and Post-processing*: The quantum state is read out through a structured, action-aligned set of Pauli- Z expectation values,

$$\hat{\mathbf{o}}_t = \left[\underbrace{\langle Z_u \rangle}_{u \in \mathcal{U}}, \underbrace{\langle Z_u Z_r \rangle}_{u \in \mathcal{U}, r \in \mathcal{R}}, \underbrace{\frac{1}{|\mathcal{U}|-1} \sum_{u' \neq u} \langle Z_u Z_{u'} \rangle}_{u \in \mathcal{U}}, \underbrace{\langle Z_r \rangle}_{r \in \mathcal{R}} \right] \in [-1, 1]^{N_Q}, \quad N_Q = (n_q - M)(M + 2) + M, \quad (21)$$

where $\mathcal{U} = \{0, \dots, n_q - M - 1\}$ and $\mathcal{R} = \{n_q - M, \dots, n_q - 1\}$ index the user and IRS qubits, respectively; the four blocks — per-user $\langle Z_u \rangle$, user-IRS $\langle Z_u Z_r \rangle$, mean user-cluster correlation, and per-IRS $\langle Z_r \rangle$ — are observables aligned with the user $\rightarrow$ IRS assignment action. Expectation-value readout is preferred to raw single-shot measurements because it returns a smooth, bounded, real-valued vector that yields stable gradient estimates and keeps the policy output well-conditioned. The single-qubit terms summarize each qubit's marginal state, while the two-qubit terms expose the pairwise correlations created by the entangling block, giving the policy access to the encoded user-IRS coupling at a cost of only N_Q action-aligned features. During training the behavior policy samples actions from $S_{\text{tr}}=1500$ -shot estimates, while the gradient Jacobian and the PPO importance ratio use the exact statevector expectations (low variance); at inference the policy uses $S_{\text{ev}}=3000$ -shot estimates.

The readout $\hat{\mathbf{o}}_t$ has a fixed length N_Q that does not match the $K(M+1)$ logits the assignment requires, and a shallow circuit exposes only a limited set of observables. A lightweight SoftmaxPQC head ξ — a single linear map followed by a trainable inverse-temperature τ , after [15] — therefore maps the quantum features to the action space (its $\sim 2\text{K}$ parameters, versus a $\sim 50\text{K}$ ReLU-MLP, underpin the parameter-efficiency result of Section V-B2). It first concatenates the readout with the raw latent into $\mathbf{h}_t = [\mathbf{z}_t \parallel \hat{\mathbf{o}}_t] \in \mathbb{R}^{2n_q + N_Q}$ as a skip connection, which preserves information that the limited readout may discard, and then outputs a per-user assignment distribution,

$$\boldsymbol{\pi}_k = \text{softmax}\left(\tau [\mathbf{W}\mathbf{h}_t + \mathbf{b}]_{k(M+1):(k+1)(M+1)}\right) \in \Delta^M, \quad k \in \mathcal{K}, \quad (22)$$

over the $M + 1$ choices (the direct link or one of the M IRS panels). The assignment is sampled $\varphi_k \sim \boldsymbol{\pi}_k$ during training and taken greedily, $\varphi_k = \arg \max \boldsymbol{\pi}_k$, at evaluation. The VQC parameters $\{\boldsymbol{\lambda}_y, \boldsymbol{\lambda}_z, \boldsymbol{\theta}^{(1)}, \dots, \boldsymbol{\theta}^{(L)}\}$ are trained by SPSA (Section IV-D), and ξ by back-propagation.

D. Entropy-Regularized Hierarchical Actor-Critic Training

The four actors of Section IV are trained together with a single shared critic in an entropy-regularized actor-critic scheme optimized by proximal policy optimization (PPO) [33]. The quantum actor and the three classical sub-actors constitute a *joint hierarchical policy* that factorizes the composite action, while the critic supplies a common advantage signal that couples their updates toward the shared goal of sum-rate maximization under per-user QoS constraints.

1) *Markov Decision Process Formulation:* We cast the sequential resource-allocation task as a Markov decision process $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma_d)$. The *state* $\mathbf{s}_t$ is the affinity feature vector of Eq. (15), built from the imperfect channel estimates $\hat{g}$. The *action* is the joint decision produced by the hierarchical policy, $\mathbf{a}_t = (\varphi, \Phi, \mathbf{w}, \mathbf{C})$, where the assignment φ is drawn from the quantum actor's distribution (22), and Φ , $\mathbf{w}$, and $\mathbf{C}$ are produced in turn by the phase, power, and common-rate sub-actors conditioned on φ . Once the action is applied, the environment advances the user positions and channels to yield the next state $\mathbf{s}_{t+1}$.

The *reward* links the optimization objective (14a) to the QoS constraint (14h) through a soft quadratic penalty,

$$r_t = R_{\text{tot}} - \lambda_D \sum_{k \in \mathcal{K}} \left(\frac{\max(0, R_k^{\min} - R_k)}{R_k^{\min} + \epsilon_q} \right)^2, \quad (23)$$

where $R_k = R_{k,p,g} + C_{k,g}$ is the total achievable rate of user k , $\lambda_D > 0$ weights the penalty, and ϵ_q is a small constant that guards the denominator. The penalty vanishes when every user meets its demand and grows quadratically with the normalized shortfall otherwise, so maximizing r_t steers the policy toward the constrained sum-rate optimum of (P0) without imposing a hard, non-differentiable feasibility boundary. To suppress the variance injected by the per-step noise, r_t is averaged over several independent noise realizations of the same action, and future rewards are discounted by a factor $\gamma_d \in (0, 1)$, distinct from the SINR γ . Here R_{tot} and the per-user rates R_k entering the reward and the QoS check are the *achieved* rates on the true channel g , whereas all four sub-policies are conditioned on the estimated channel $\hat{g}$; the penalty thus charges the policy for the design-versus-reality gap.

2) *Learning Objective and Critic:* The hierarchical policy is trained to maximize an entropy-regularized expected return,

$$\mathcal{J}(\Theta) = \mathbb{E} \left[\sum_t \gamma_d^t \left(r_t + \beta \sum_j H(\pi_j(\cdot | \mathbf{s}_t)) \right) \right], \quad (24)$$

where Θ collects the parameters of all four actors, $H(\pi_j) = -\sum \pi_j \log \pi_j$ is the entropy of actor j 's policy, and $\beta > 0$ weights an entropy bonus that sustains exploration over the combinatorial assignment space and prevents premature convergence to a deterministic policy. The coefficient β is annealed linearly from β_0 to a floor $\beta_{\min}$ over the early training phase, so that exploration dominates initially and gradually gives way to exploitation. The reward term of (24) drives the policy toward the constrained sum-rate optimum, while the entropy term keeps the search broad.

Optimizing (24) by policy gradient requires an estimate of how advantageous each action is relative to the current policy; this is supplied by a shared critic. The critic $V(\mathbf{s}_t; \psi)$ is an MLP that estimates the state value and acts as a variance-reducing baseline. The advantage shared by all actors is obtained by generalized advantage estimation [35],

$$\hat{A}_t = \sum_{l \geq 0} (\gamma_d \lambda_{\text{GAE}})^l \delta_{t+l}, \quad \delta_t = r_t + \gamma_d V(\mathbf{s}_{t+1}; \psi) - V(\mathbf{s}_t; \psi), \quad (25)$$

where $\lambda_{\text{GAE}} \in [0, 1]$ trades bias against variance. Conditioning the baseline on the state alone makes $\hat{A}_t$ a low-variance signal common to the whole hierarchy and avoids the gradient self-cancellation of a state-action baseline. Advantages are standardized across each rollout batch before use. The critic's parameters are fitted by regressing toward the resulting λ -return target $G_t = \hat{A}_t + V(\mathbf{s}_t; \psi)$,

$$\mathcal{L}_V(\psi) = (G_t - V(\mathbf{s}_t; \psi))^2, \quad (26)$$

with $V(\mathbf{s}_t; \psi)$ further stabilized by PopArt value-target normalization [36]: the network regresses a running-mean/variance-normalized target and its output layer is rescaled at every statistics update to preserve the real-valued prediction, decoupling the learning rate from the reward scale, which otherwise shifts markedly across the R_{LoS} curriculum stages. No separate target network or Polyak averaging is used; the multi-step GAE bootstrap already yields a sufficiently low-variance regression target given the entropy-regularized, clipped PPO updates.

3) *Policy Optimization via PPO*: Let $j \in \{\varphi, \Phi, \mathbf{w}, \mathbf{C}\}$ index the four actors with policies $\pi_j(\cdot | \mathbf{s}_t; \Theta_j)$, and let $\rho_t^j = \pi_j(a_t^j | \mathbf{s}_t) / \pi^{j, \text{old}}(a_t^j | \mathbf{s}_t)$ denote the importance ratio between the current and behavior policies.

Every actor is updated with the clipped PPO surrogate

$$\mathcal{L}_{\text{PPO}}^j = \mathbb{E} \left[\min(\rho_t^j \hat{A}_t, \text{clip}(\rho_t^j, 1 - \epsilon_c, 1 + \epsilon_c) \hat{A}_t) \right], \quad (27)$$

where ϵ_c is the clip range. Clipping bounds each update so that a large change in the high-level assignment cannot abruptly shift the conditioning of the downstream actors, and it permits several optimization epochs over the same batch. Combining the clipped surrogate with the entropy bonus of the objective in (24) and, for the quantum actor, the autoencoder reconstruction loss (18), the per-actor training objectives become

$$\begin{aligned} \mathcal{J}_\varphi &= \mathcal{L}_{\text{PPO}}^\varphi + \beta H(\pi_\varphi) - \alpha_{AE} \mathcal{L}_{AE}, \\ \mathcal{J}_j &= \mathcal{L}_{\text{PPO}}^j + \beta H(\pi_j), \quad j \in \{\Phi, \mathbf{w}, \mathbf{C}\}. \end{aligned} \quad (28)$$

The quantum encoding scales and variational angles $\boldsymbol{\eta} = \{\boldsymbol{\lambda}_y, \boldsymbol{\lambda}_z, \boldsymbol{\theta}\}$ lie inside the circuit and cannot be differentiated by ordinary back-propagation; their gradients are instead estimated by simultaneous perturbation stochastic approximation (SPSA) [34]. Whereas the parameter-shift rule evaluates two circuits *per* parameter — a cost that scales linearly with $\dim(\boldsymbol{\eta})$ — SPSA perturbs all parameters at once along a single random direction $\boldsymbol{\Delta}$ with $\Delta_i \in \{-1, +1\}$ and estimates each component from one pair of circuit evaluations,

$$\hat{J}_{o,i} = \frac{\hat{o}_o(\boldsymbol{\eta} + c_s \boldsymbol{\Delta}) - \hat{o}_o(\boldsymbol{\eta} - c_s \boldsymbol{\Delta})}{2 c_s} \Delta_i, \quad \frac{\partial \mathcal{J}_\varphi}{\partial \eta_i} = \sum_o \frac{\partial \mathcal{J}_\varphi}{\partial \hat{o}_o} \hat{J}_{o,i}, \quad (29)$$

where c_s is a small perturbation step: SPSA estimates the Jacobian of the circuit readout $\partial \hat{o}_t / \partial \boldsymbol{\eta}$ from a single pair of circuit evaluations per Rademacher direction $\boldsymbol{\Delta}$ — a cost independent of $\dim(\boldsymbol{\eta})$ — and averages the estimate over a few repetitions to reduce its variance; the head, entropy, and autoencoder-reconstruction gradients are then obtained by exact back-propagation through $\partial \mathcal{J}_\varphi / \partial \hat{o}_t$. This requires roughly $6\times$ fewer circuit evaluations per update than the parameter-shift rule; the price is a higher-variance gradient, which the critic baseline and the conservative PPO clipping are well placed to absorb. The classical encoder and post-processing head of the quantum actor, all three sub-actors, and the critic are updated by standard back-propagation, and every module uses the Adam optimizer with per-module learning rates and gradient-norm clipping.

4) Training Procedure: Training proceeds through repeated rollout-and-update cycles. *(i) Rollout:* the current policy is executed for a fixed number of episodes, and each transition $(\mathbf{s}_t, \mathbf{a}_t, r_t, V_t)$ is stored

together with the behavior log-probabilities $\log \pi^{j,\text{old}}$ of all four actors. (ii) *Targets*: the stored values and the next-state bootstrap are used to compute the GAE advantages and λ -return targets (25)–(26), after which the advantages are standardized over the batch. (iii) *Update*: for several epochs the buffer is shuffled into mini-batches and, for each mini-batch, the four actor objectives (28) and the critic loss (26) are evaluated and their parameters updated; the importance ratios in (27) make these repeated passes both stable and sample-efficient. (iv) *Anneal*: the PopArt value-normalization statistics are refreshed and the entropy coefficient and learning rates are decayed on their schedules. The autoencoder warm-up of Section IV-B is performed once before this loop begins, and the cycle repeats until convergence. The hyperparameter values used in our experiments are reported in Section V.

V. EXPERIMENT

A. Simulation Setup

We evaluate the proposed HQC-HAC framework on the IRS-assisted grouped RSMA downlink of Section III: a LEO satellite (1.58 GHz, 800 km) serves K single-antenna users aided by M rooftop-mounted IRS panels of $N = 24$ reflecting elements with 2-bit phase quantization. In each episode the panels and users are spawned within the LoS coverage disk, with a fraction of users placed inside building footprints where the direct link is blocked and only an IRS can restore coverage — the worst-user bottleneck the framework must resolve. Users then move by random walk at 1.5 m/s while the three ITU-R links, rain attenuation, and additive noise are re-sampled at every step, and the agent observes only imperfect CSI ($\kappa = 0.05$). Each episode runs 200 steps under the per-user demand $R_k^{\min} = 0.1$ bps/Hz. Three topologies scale the problem: $(K, M, P_S) = (5, 1, 50 \text{ dBm})$, $(10, 2, 50 \text{ dBm})$, and $(15, 3, 50 \text{ dBm})$, with VQC depth $L = 2/3/5$, respectively; all remaining settings are listed in Table III.

Training protocol. Training follows Section IV-D under a blockage curriculum: the LoS radius R_{LoS} is raised from 0.2 to 0.5 km in stages of 2000 episodes, each stage warm-starting from the previous one (dotted lines in Figs. 6–7). Before each update, 12 rollout episodes are collected; the four actors and the critic are then optimized for 6 PPO epochs on mini-batches of 64 transitions, with the entropy coefficient annealed over the first 10% of episodes. Quantum parameters are updated by SPSA and all classical modules by Adam, with per-module gradient-norm clipping; the autoencoder warm-up precedes joint training.

TABLE III: Simulation and Training Hyperparameters

Parameter	Value	Parameter	Value
<i>System, channel, and quantum actor</i>		<i>Reinforcement learning and training</i>	
Users K / IRS M (Case 1/2/3)	(5, 1) / (10, 2) / (15, 3)	Discount factor γ_d	0.95
Elements per IRS N	24	GAE parameter λ_{GAE}	0.9
Satellite power P_S (Case 1/2/3)	50 / 50 / 50 dBm	PPO clip ϵ_c / epochs	0.2 / 6
Carrier frequency f / altitude h_{SR}	1.58 GHz / 800 km	Rollout episodes / mini-batch	12 / 64
Antenna gains G_S, G_U	60 dBi	Entropy $\beta_0 \rightarrow \beta_{\min}$	$10^{-3} \rightarrow 3 \times 10^{-4}$
Noise n_0 (mean, variance)	(−30, 10) dBW	QoS penalty λ_D / guard ϵ_q	$1.5 / 10^{-3}$
Rain attenuation (μ, σ^2)	(4 dB, 0.1)	Reward noise averaging	16
CSI error coefficient κ	0.05	LR (AE ω / VQC η / head ξ)	$10^{-4} / 10^{-4} / 3 \times 10^{-4}$
Phase quantization bits B	2 (4 levels)	LR (phase, power / C_k / critic)	$10^{-4} / 5 \times 10^{-5} / 3 \times 10^{-5}$
QoS demand $R_k^{\min}$	0.1 bps/Hz	Episodes per curriculum stage / steps	2000 / 200
LoS radius R_{LoS} / IRS height	0.2–0.5 km (curriculum) / 20 m	Optimizer	Adam
User speed	1.5 m/s	Gradient-norm clip (AE / actors)	0.1 / 0.5
Qubits n_q / latent dim $2n_q$	12 / 24		
Variational layers L (Case 1/2/3)	2 / 3 / 5		
Data re-uploading	enabled		
SPSA step c_s / repetitions	0.1 / 16		
Measurement shots S (train / eval)	1500 / 3000		
Autoencoder loss weight α_{AE}	0.5		

Evaluation protocol and baselines. The learned policy is executed greedily ($\varphi_k = \arg \max \pi_k$, 3000 shots) over 50 held-out episodes of 200 steps with a fixed evaluation seed. We report the system sum-rate R_{tot} as the primary metric, together with the QoS-satisfaction rate — the fraction of users meeting $R_k^{\min}$. HQC-HAC is compared against four baselines sharing the same grouped-RSMA signal model and differing only in the assignment policy: (i) *greedy*, which selects for each user the strongest estimated link; (ii) *direct-only*, which disables the IRS; (iii) *all-IRS*, which routes every user through an IRS; and (iv) *random* assignment. Robustness to CSI error and receiver noise is assessed zero-shot, re-evaluating the trained policy under test-time κ and σ^2 overrides without retraining.

B. Performance Analysis

1) Sum-Rate and QoS Against Baselines: Figs. 6–7 show the training trajectories across the blockage curriculum, and Tables IV–V the corresponding greedy-inference results. The learned assignment outperforms every fixed baseline on both metrics: HQC-HAC exceeds the strongest baseline (Greedy) by 19.0% in sum-rate in Case 1 and 11.3% in Case 2, at equal or better QoS, and the QoS separation widens with scale (+0.8 pp at $K = 5$ versus +5.3 pp at $K = 10$). The blind policies illustrate why the assignment

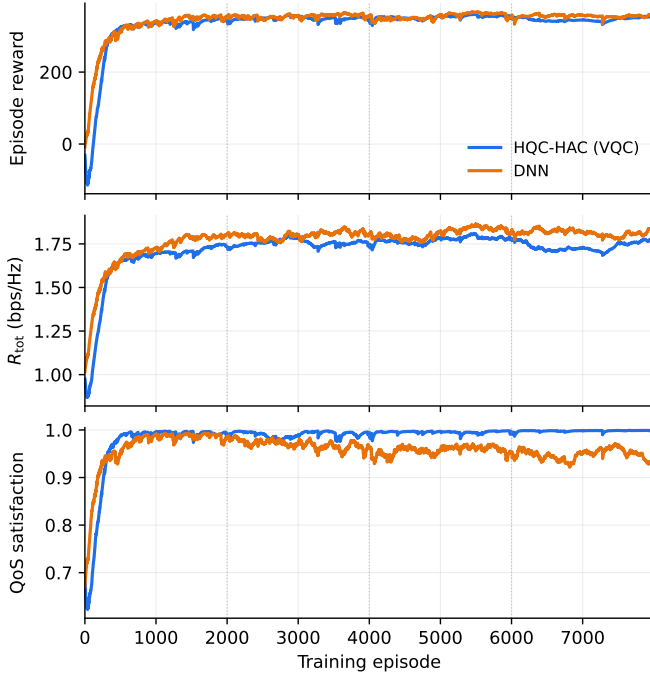


Fig. 6: Case 1 ($K=5$, $M=1$) training trajectories of HQC-HAC (VQC) vs. the classical DNN actor over the full R_{LoS} curriculum ($0.2 \rightarrow 0.5$); dotted lines mark ramp transitions.

TABLE IV: Case 1 ($K = 5$, $M = 1$, $P_S = 50$ dBm): greedy-inference comparison against base-lines. VQC and DNN each at its sweet-spot checkpoint (ep 1900 and ep 1100, respectively); 50 episodes $\times$ 200 steps, seed 42, eval env $R_{\text{LoS}} = 0.5$ (full-unlock spawn geometry).

Policy	Reward	R_{tot}	QoS
HQC-HAC	354.68	1.7748	99.8%
DNN	359.98	1.8196	94.9%
Greedy	295.42	1.5062	98.3%
All-IRS	-415.67	0.7304	57.6%
Direct-only	-129.24	0.8345	51.3%
Random	-448.93	0.5551	30.2%

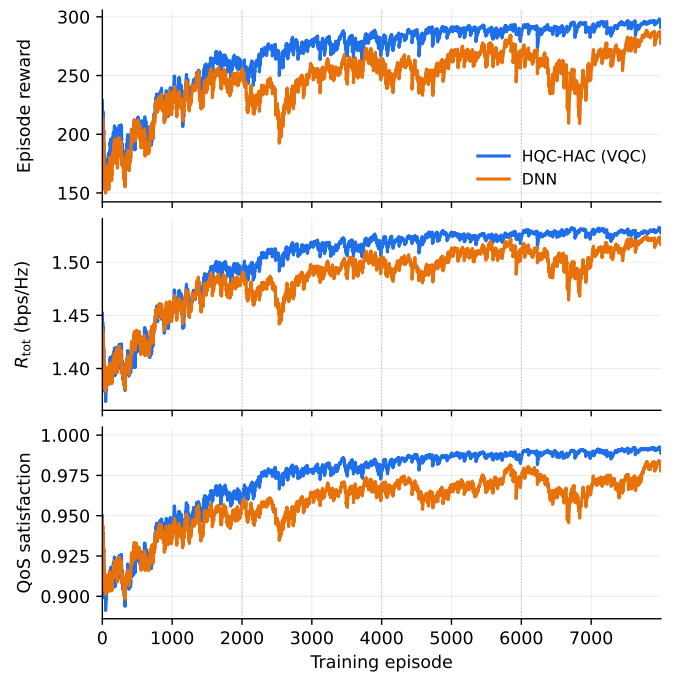


Fig. 7: Case 2 ($K=10$, $M=2$) training trajectories over the full curriculum ($0.2 \rightarrow 0.5$); dotted lines mark ramp transitions.

TABLE V: Case 2 ($K = 10$, $M = 2$, $P_S = 50$ dBm): greedy-inference comparison.

Policy	Reward	R_{tot}	QoS
HQC-HAC	294.91	1.5300	97.3%
DNN	280.03	1.5181	95.4%
Greedy	215.28	1.3752	92.0%
All-IRS	-1852.63	0.3871	27.2%
Direct-only	-990.09	0.6867	39.4%
Random	-1938.94	0.3037	7.6%

is the decisive variable: Direct-only strands the blocked users, All-IRS strands those with a strong direct link, and both — like Random — collapse to 7.6–59.0% QoS with large negative rewards, whereas the learned policy sustains $\geq 97\%$ QoS while serving all K users.

2) *Quantum versus Classical Actor: Parameter Efficiency and the Pareto Frontier:* To isolate the contribution of the quantum actor, the VQC block is replaced by a classical MLP of identical role, input, and output, trained under the same recipe (Tables IV–V, row DNN). The two actors reach sum-rates within 3% of each other but settle on two distinct operating points of the same trade-off: the VQC leans toward

TABLE VI: Assignment-actor parameter count: VQC SoftmaxPQC vs. classical MLP (phase/power/ C_k sub-actors identical in both).

Actor	Case 1	Case 2	Case 3
HQC-HAC (VQC)	120	144	(pending)
DNN (MLP)	40,586	43,166	(pending)
Ratio DNN/VQC	338×	300×	(pending)

QoS (+4.9 pp in Case 1, +1.9 pp in Case 2), the DNN toward raw sum-rate, so neither dominates the other — consistent with the prior that parametrized quantum policies match, rather than exceed, classical ones in task performance [15]. The separation is in parameter efficiency: the VQC assignment actor attains this operating point with 120/144 trainable parameters against 40,586/43,166 for the MLP (Table VI) — a 338×/300× reduction — with the VQC count growing only linearly in the register size (Appendix D). In Case 2 the VQC additionally traverses the curriculum transitions with visibly smaller transient drops (Fig. 7).

3) *Robustness to System Scale and Channel Conditions:* Table VII varies the reflecting-element count: across $N = 20$ –36 the achievable sum-rate stays within 1.71–1.75 bps/Hz at $\geq 99.4\%$ QoS, with no monotone trend in N . The passive aperture gain benefits the Greedy baseline identically, so the agent’s margin remains at +13–17% throughout: at $P_S = 50$ dBm the system is power- rather than aperture-limited, and enlarging the IRS is not the binding lever. Table VIII instead sweeps the satellite transmit power over $P_S = 40$ –70 dBm: the sum-rate climbs monotonically from 1.41 to 2.45 bps/Hz, while QoS saturates near 99.7% for $P_S \geq 50$ dBm and drops to 91.6% at 40 dBm as the link turns power-limited. The same sweep delimits the value of the IRS — at $P_S = 70$ dBm the direct link alone already serves every user (Direct-only QoS 100%), rendering the reflecting surface redundant, whereas at 40 dBm Direct-only satisfies only 41.5% of users and the IRS becomes the decisive resource. Table IX evaluates the trained Case-2 policy zero-shot under degraded channel knowledge: quadrupling the CSI error ($\kappa = 0.05 \rightarrow 0.20$) costs only 0.3 pp QoS and 0.2% sum-rate, and raising the receiver-noise variance from 10 to 16 dB² costs 1.0 pp QoS, while the lead over Greedy (≈ 5 pp QoS) is preserved at every noise level — the policy degrades gracefully without retraining.

TABLE VII: Comparison of the achievable sum-rate under various IRS-element counts N on $K = 5, M = 1$

N	R_{tot}	QoS	Margin vs. Greedy
20	1.7141	99.4%	+16.5%
24	1.7465	99.7%	+16.0%
28	1.7483	99.8%	+14.8%
32	1.7064	99.5%	+13.2%
36	1.7529	99.6%	+15.8%

TABLE VIII: Comparison of the achievable sum-rate under various satellite transmit power levels P_S on $K = 5, M = 1$

P_S (dBm)	R_{tot}	QoS
40	1.4065	91.6%
50	1.7465	99.7%
60	1.8560	99.9%
70	2.4518	99.5%

VI. DISCUSSION

The results position HQC-HAC as a *feasibility and efficiency* contribution rather than a claim of quantum task-advantage. On the multi-IRS grouped-RSMA assignment problem the variational quantum actor reaches the sum-rate and QoS of a classical deep-network actor to within 3% (Section V-B2) using two orders of magnitude fewer parameters, and — unlike the per-state convex-optimization pipeline of Tan et al. [8] — amortizes its cost into a single offline training run, after which each channel realization is served by one forward pass. This is the honest reading advocated throughout: parametrized quantum policies match, rather than exceed, classical ones on natural control tasks [15], so the value here lies in the parameter budget and amortized deployment, not in the achievable objective.

The ablations delineate which design choices carry this result and which are inert. The autoencoder is a genuine *scaling enabler*: without it the affinity vector would demand 22 and 41 qubits at Cases 2 and 3 (Table XIV), placing the state-vector simulation with per-step SPSA gradients out of reach, whereas the fixed 12-qubit register keeps the problem tractable at a reconstruction cost that grows only with the compression ratio (Fig. 10). Two ingredients that are load-bearing in other quantum-policy settings are, by contrast, *removable* here: the trainable encoding scales λ can be left at their initial values at no measurable cost — not because the scales are irrelevant, but because the surrounding network co-adapts so that the frozen value already sits at the optimum of the rescaling direction (Appendix C) — and the readout’s

TABLE IX: Comparison of the achievable sum-rate under various CSI error coefficients κ and receiver-noise variances σ^2 on $K = 10, M = 2$ (zero-shot, no retraining)

Noise level	R_{tot}	QoS
base ($\kappa=0.05$)	1.5300	97.3%
$\kappa = 0.10$	1.5291	97.2%
$\kappa = 0.20$	1.5263	97.0%
$\sigma^2 = 13$	1.5270	96.8%
$\sigma^2 = 16$	1.5241	96.3%

internal structure is not the performance bottleneck. The hierarchical decomposition is decisive: an oracle-intervention ladder places assignment far above phase, power, and common-rate as the dominant lever, which the sequential factorization isolates cleanly.

The system-level sweeps clarify the operating regime. The agent’s margin over the strongest baseline is invariant to the reflecting-element count N at fixed power, indicating a power- rather than aperture-limited regime; the transmit-power sweep makes this explicit, with the IRS contribution vanishing once the direct link alone satisfies every user at high P_S . The trained policy also degrades gracefully under zero-shot CSI-error and receiver-noise perturbations, retaining its lead without retraining.

Several limitations qualify these findings. The reported comparisons use a single training seed and a fixed classical configuration; a multi-seed, iso-performance grid would tighten the parameter-efficiency claim. The largest case ($K=15$) is characterized only structurally, and the depth, entangling-bridge, and larger-scale readout ablations remain to be run. Finally, the inference cost is an algorithmic per-state complexity measured in simulation, not a wall-clock result on quantum hardware.

VII. CONCLUSION

We presented HQC-HAC, a hybrid quantum–classical hierarchical actor–critic for sum-rate maximization in IRS-assisted grouped-RSMA integrated satellite–terrestrial networks. The formulation casts the joint assignment, phase, power, and common-rate problem as a sequential dependency chain under QoS, discrete-phase, blockage, and imperfect-CSI constraints; a variational quantum circuit resolves the combinatorial $(M+1)^K$ assignment while three classical sub-actors handle the factorable low-level controls, and an autoencoder compresses the state into a NISQ-sized register. Across two problem scales the method exceeds the strongest heuristic baselines and matches a classical deep-network actor to within 3% using $338 \times / 300 \times$ fewer assignment-actor parameters, while amortizing into a single offline training run the per-

state solve cost of prior convex-optimization approaches. Ablations attribute the result to the hierarchical decomposition and the autoencoder compression and honestly identify the encoding scales and readout structure as removable degrees of freedom in this task. Future work includes the full $K=15$ case, multi-seed and iso-performance comparisons, the remaining architectural sweeps, and evaluation on quantum hardware.

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APPENDIX A

GROUP-COMPOSITION CAPACITY OF AN IRS

Setup. Consider one IRS serving two user groups: group 0 (direct transmission) with A users and group 1 (IRS-assisted transmission) with B users. We analyze the incremental effect of adding one free user to the IRS group, i.e., re-assigning a user from group 0 to group 1. The power allocation and beamforming weights are assumed fixed throughout the re-assignment, so that the resulting ΔR_{tot} isolates the pure effect of group composition. User rates are expressed as functions of the effective SNR, $R_p(\Gamma)$ and $R_c(\Gamma)$, where $\Gamma = |h|^2 P_S / \sigma^2$:

$$\Delta R_{\text{tot}}(a) = \underbrace{R_p(\Gamma_a^{\text{IRS}}) - R_p(\Gamma_a^{\text{dir}})}_{\text{(I) private re-route}} + \underbrace{R_{c,A}^{\text{new}} - R_{c,A}^{\text{old}}}_{\text{(II) direct-group relief}} + \underbrace{\min(R_{c,B}^{\text{old}}, R_c(\Gamma_a^{\text{IRS}})) - R_{c,B}^{\text{old}}}_{\text{(III) common-rate degradation}} \quad (30)$$

- **(I):** a 's private rate on the IRS route (after) minus on the direct route (before).
 - > 0 : the reflected path offers the higher private rate (the N^2 cascade gain outweighs the direct link), favoring re-assignment. For a *blocked* user $\Gamma_a^{\text{dir}} \approx 0$, so (I) is always dominant-positive: routing blocked users to their IRS is monotonically beneficial, with no threshold in A .

- < 0 : the direct link offers the higher private rate; the re-assignment loses private rate and must be justified by (II).
- **(II):** the direct-group common rate after minus before a leaves.
 - $= 0$: a was not the minimum-SINR member of group 0, so the group minimum — and hence its common rate — is unchanged.
 - > 0 : a was the worst — the common rises to the next-worst member.
 - < 0 : only when a is the *last* member of group 0: the group vanishes and its common stream is forfeited, $(II) = -R_c(\Gamma_{\text{dir}})$ (specific to fixed power weights; predicted 0.141 bps/Hz against 0.137 simulated at $P_S=50$ dBm).
- **(III):** the IRS-group common rate after minus before a joins. Adding a member can only keep or lower a min-SINR common rate, so $(III) \leq 0$.
 - $= 0$: the admitted user does not become the minimum-SINR member of the IRS group; the common rate is unchanged.
 - < 0 : the admitted user becomes the new group minimum; the common rate drops to $R_c(\Gamma_a^{\text{IRS}})$, reducing the rate of every member of the group simultaneously.

Marginal of admitting free users. Under representative two-point SNRs ($\Gamma_B < \Gamma_A$ deterministic), the first admission carries a one-time common-rate reduction, and every later admission is neutral to the group minimum:

$$\Delta_{0 \rightarrow 1} = [R_p(\Gamma_B) - R_p(\Gamma_{\text{dir}})] + \underbrace{[R_c(\Gamma_B) - R_c(\Gamma_A)]}_{\text{one-time common-rate reduction}}. \quad (31)$$

Under per-user fading, however, neither the single reduction nor the absence of later ones is guaranteed — both are order-statistics events of the group minimum. In expectation,

$$E[\Delta_{B \rightarrow B+1}] = \underbrace{E[R_p(\Gamma_{B+1})] - R_p(\Gamma_{\text{dir}})}_{\text{private (own fading realization)}} + \underbrace{E[R_c(m_{B+1})] - E[R_c(m_B)]}_{<0, \downarrow 0 \text{ (common-rate reduction)}}, \quad (32)$$

where m_B denotes the minimum effective SNR among the $A+B$ group members (characterized below). The reduction term is negative but vanishes as B grows, so $E[\Delta_{B \rightarrow B+1}]$ *increases* with B : under i.i.d. member fading the cumulative sum-rate is convex in B and its optimum lies at a boundary ($B=0$ or

the maximal B); an interior optimum arises only when the candidates are heterogeneous and admitted in decreasing order of mean cascade SNR. The sum-rate carrying capacity is

$$B_{\text{rate}}^*(A) = \arg \max_B R_{\text{tot}}(A, B). \quad (33)$$

Order statistics of the group minimum. Under Rayleigh fading of the IRS–user link, the cascade SNR of member j factorizes into a deterministic mean (set by the geometry) and a unit exponential draw:

$$\Gamma_j = \bar{\Gamma}_j X_j, \quad X_j \sim \text{Exp}(1) \text{ i.i.d.}, \quad (34)$$

so the group minimum is itself exponential — exactly, for heterogeneous means:

$$m \sim \text{Exp}(\Lambda), \quad \Lambda = \sum_{i \in A} \frac{1}{\bar{\Gamma}_{A,i}} + \sum_{j \in B} \frac{1}{\bar{\Gamma}_{B,j}}, \quad E[m] = \frac{1}{\Lambda}, \quad P(m = \Gamma_k) = \frac{1/\bar{\Gamma}_k}{\Lambda}, \quad (35)$$

and every expectation above admits a closed form via the exponential integral E_1 :

$$\mathbb{E}[\log_2(1 + cX)] = \frac{e^{\lambda/c} E_1(\lambda/c)}{\ln 2}, \quad X \sim \text{Exp}(\lambda). \quad (36)$$

The probability that an admitted user becomes the group minimum — i.e., that the common rate is reduced — is, for the first admission,

$$P(\Gamma_b < \min_{i \in A} \Gamma_i) = \frac{1/\bar{\Gamma}_B}{1/\bar{\Gamma}_B + A/\bar{\Gamma}_A} = \frac{\bar{\Gamma}_A}{\bar{\Gamma}_A + A\bar{\Gamma}_B} < 1 \quad (37)$$

and for the $(B+1)$ -th admission,

$$P(\Gamma_b < \min_{i \in A, B} \Gamma_i) = \frac{1/\bar{\Gamma}_B}{(B+1)/\bar{\Gamma}_B + A/\bar{\Gamma}_A} \xrightarrow{\bar{\Gamma}_A \gg \bar{\Gamma}_B} \frac{1}{B+1}. \quad (38)$$

QoS bound. The common-rate shares C_k are decision variables, so a group is feasible iff its common budget covers the total shortfall $\varsigma_k = (R_k^{\min} - R_p(\Gamma_k))^+$ — exact, with no assumption on how the common rate is shared:

TABLE X: Composition thresholds in a controlled experiment ($K=10$, $N=24$; one IRS deactivated and excluded from grouping and power allocation, power weights fixed, $A=3$ blocked users on the remaining IRS, free users admitted in decreasing order of mean cascade SNR; 50 independent spawns per power level, all B evaluated on common channel realizations).

Quantity	$P_S = 50$ dBm	$P_S = 40$ dBm
$\max R_{\text{tot}}^{\text{theory}} - R_{\text{tot}}^{\text{sim}} $	6.7×10^{-16}	8.9×10^{-16}
Mean marginal $\Delta_{0 \rightarrow 1}, \Delta_{1 \rightarrow 2}$ (bps/Hz)	$-0.061, -0.108$	$-0.090, -0.104$
B_{rate}^* (mode / mean)	0 / 0.34	0 / 0.32
B_{QoS}^* , general (mode / mean)	2 / 1.44	0 / 0.54
B_{QoS}^* , equal share (mode / mean)	1 / 1.04	0 / 0.42
$B^* = \min(\cdot, \cdot)$ (mode / mean)	0 / 0.34	0 / 0.32
Per-draw prediction accuracy (all four thresholds)	100%	100%

$$B_{\text{QoS}}^*(A) = \max \left\{ B : \sum_{k=1}^{A+B} \varsigma_k \leq R_c(\Gamma_{\min}(B)) \right\} \quad (39)$$

(note the double effect of B : the shortfall mass grows while the budget $R_c(\Gamma_{\min}(B))$ shrinks, since the minimum is taken over more members). Equal sharing ($C_k = R_c/n$) yields the conservative special case

$$B_{\text{QoS}}^* = \lfloor n_{\text{mix}}^* \rfloor - A, \quad n_{\text{mix}}^* = \frac{R_c(\Gamma_{\min})}{R_k^{\min} - R_p(\Gamma_{\text{wk}})}, \quad (40)$$

which coincides with the general bound when the shortfalls are homogeneous and was verified exactly in 1066/1066 simulated (A, B) cells. The binding composition threshold combines both sides:

$$\boxed{B^*(A) = \min(B_{\text{rate}}^*, B_{\text{QoS}}^*)} \quad (41)$$

Table X summarizes the resulting thresholds. The first row shows that the closed forms above reproduce the simulated sum-rate exactly (deviations at machine precision). The admission marginals (second row) form a decreasing sequence, as predicted for heterogeneous candidates admitted in decreasing order of cascade SNR: each next user contributes a smaller private gain and pulls the group minimum lower. The two QoS rows show the general shortfall-based bound admitting more users than its equal-share special case (2 vs. 1 at $P_S=50$ dBm), confirming the latter is conservative, while the binding B^* (second-to-last row) is set by the sum-rate side in this geometry and is therefore insensitive to the sharing assumption. The last row reports that all four thresholds are predicted correctly on every individual realization.

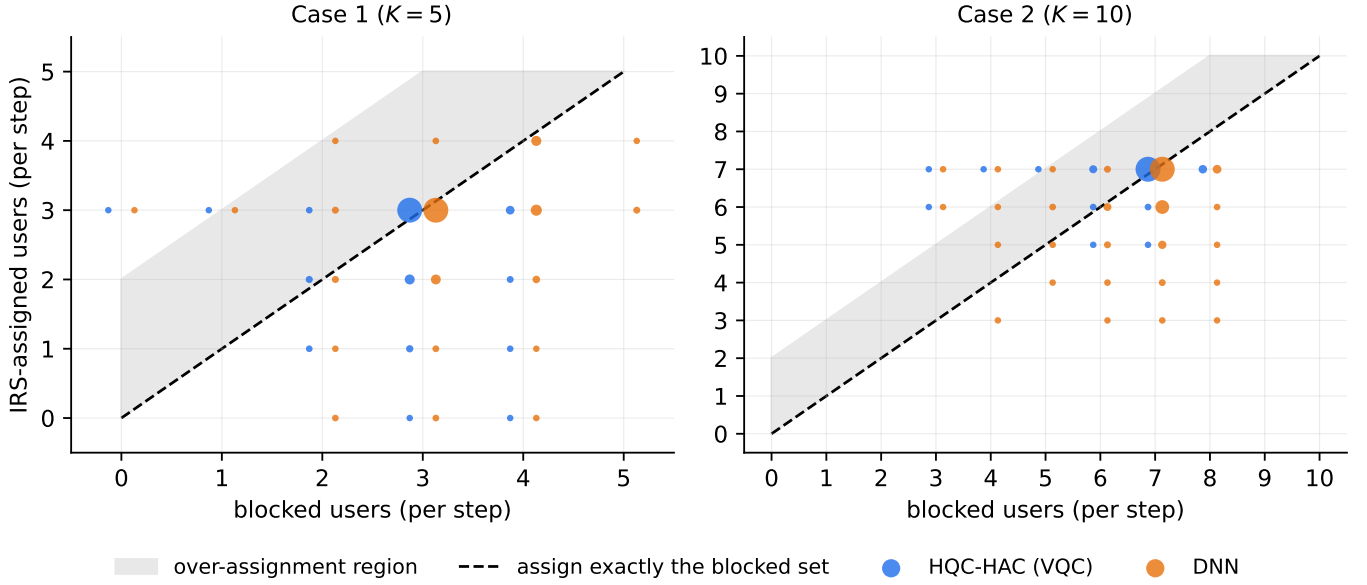


Fig. 8: Per-step assignment of the trained policies against the composition rule (dashed line: assign exactly the blocked set; marker size $\propto$ frequency over 25 greedy episodes on the checkpoints of Tables IV–V). The shaded band marks the over-assignment region, where the admission marginal of Eq. (30) is near zero.

Trained-policy check. Fig. 8 closes the loop from the theory to the deployed agents: at every greedy-evaluation step, both actors assign almost exactly the blocked set — 99.4–100% of routed blocked users go to their own IRS, and over-assignment occurs in at most 3.8% of steps, inside the region where the admission marginal is near zero. The VQC tracks the rule more tightly than the DNN in Case 2 (mean deviation $|\Sigma_{\text{IRS}} - \#\text{blocked}| = 0.10$ vs. 0.35; transient under-assignment in 5.5% vs. 27.1% of steps), an assignment-level counterpart of its QoS edge in Section V-B2.

APPENDIX B

RATE-QoS PARETO FRONTIER OF THE REWARD

Definition and protocol. This appendix characterizes, independently of any learned policy, the Rate-QoS trade-off that the reward of Section IV-D encodes. For a fixed environment define the *frontier*

$$F(q) = \max_{\varphi, \Phi, \mathbf{w}, \mathbf{C}} \mathbb{E}[R_{\text{tot}}] \quad \text{s.t.} \quad \text{QoS} \geq q, \quad (42)$$

the largest expected sum-rate achievable at QoS level q under the constraints of (P0); the reward-optimal policy operates at the point of this frontier selected by the penalty weight λ_D . We evaluate F by computing per-state optima over all four decision variables on the simulator’s own channel draws (500 states per case;

assignment by exhaustive search or local search, phase exact, power on a grid — hence achievable lower bounds). The power split is generalized to arbitrary fractions $x_k = w_{p,k}/P_S$, $y_g = w_c^g/P_S$, $\sum_k x_k + \sum_g y_g \leq 1$, giving the per-user SINRs

$$\text{SINR}_{p,k} = \frac{x_k \Gamma_k}{\Gamma_k(1 - x_k - y_{g(k)}) + 1}, \quad \text{SINR}_{c,k} = \frac{y_{g(k)} \Gamma_k}{\Gamma_k(1 - y_{g(k)}) + 1}. \quad (43)$$

High-power limit (interference-limited invariance). As $\Gamma_k \rightarrow \infty$ — the $P_S=50$ dBm operating regime — the two SINRs converge to $x_k/(1 - x_k - y_{g(k)})$ and $y_{g(k)}/(1 - y_{g(k)})$: the channel magnitudes cancel, and every rate becomes a function of the power fractions alone. In this regime the frontier is set by *counting* — how many users are served and how the unit power budget is split — which is why the element count N is inert at the operating point (Table VII) and why the closed forms below require no channel statistics beyond $\Gamma_{\max}$.

Rate-maximal endpoint. Concentrating the entire budget on the single strongest user removes all intra-system interference — the only configuration that escapes the interference-limited regime, since any configuration with two or more streams is bounded by the concentration identity below:

$$V_{\text{rate}}^* = \mathbb{E}[\log_2(1 + \Gamma_{\max})], \quad \Gamma_{\max} = \max_k \max_{\text{link}} |h_k|^2 P_S / \sigma^2. \quad (44)$$

Serve-all plateau. In the saturated-SINR limit, the private sum-rate under an equal split across m served users is

$$R(m) = m \log_2 \frac{m}{m-1} \xrightarrow{m \rightarrow \infty} \frac{1}{\ln 2} \approx 1.443, \quad (45)$$

a decreasing function of m : serving everyone costs sum-rate, with a floor of $1/\ln 2$ bps/Hz. For Case 2, $R(10) = 1.520$ plus the common streams (≈ 0.09) reproduces the serve-all frontier level 1.61 to three digits.

Critical penalty weight. Under the quadratic QoS penalty, fully withholding service from one user costs $\lambda_D \hat{p}$ with $\hat{p} = (R_k^{\min}/(R_k^{\min} + \epsilon_q))^2 \approx 0.98$, so the reward-optimal number of served users follows the piecewise maximum

$$V^*(\lambda_D) = \max_{m \in \{1..K\}} \{R^*(m) - \lambda_D (K - m) \hat{p}\}, \quad \boxed{\lambda_{\text{crit}} = \frac{V_{\text{rate}}^* - R^*(K)}{(K - 1) \hat{p}}} \quad (46)$$

where $R^*(m)$ is the best sum-rate serving m users and λ_{crit} is the penalty weight above which the serve-all optimum takes over; it follows by equating the two endpoint arms of the piecewise maximum — the single-user arm $R^*(1) = V_{\text{rate}}^*$ carrying the full parking penalty $\lambda_D (K-1) \hat{p}$, the serve-all arm $R^*(K)$ carrying none.

Why 100% QoS is not the reward optimum. The penalty's marginal cost $2\lambda_D t / (R_k^{\min} + \epsilon_q)^2$ vanishes as the shortfall $t \rightarrow 0$, so the reward-optimal common-rate allocation is a water-filling solution — the shares solve $\sum_{k \in \mathcal{K}_g} (\varsigma_k - t^*)^+ = R_{c,g}$ for a common level t^* , exact for the quadratic penalty — that settles the weakest users marginally *below* their demand: even $\lambda_D \rightarrow \infty$ attains only $\sim 92\%$ counted QoS under per-step noise. Reward optimality and 100% QoS are therefore distinct targets, and we compare policies on the frontier at *matched* QoS.

Table XI summarizes the numerical frontier, evaluated in the same environments as Tables IV–V. The closed form (44) matches the numerical oracle to four digits. The critical weights place the two cases on *opposite sides* of the training value $\lambda_D = 1.5$: in Case 1 the reward's global optimum is the degenerate single-user concentration (the curriculum and entropy regularization steer the learned policy to the serve-all region instead), while in Case 2 the reward itself already prefers serving all users. The frontier shapes also differ — Case 1 satisfies $R_p(\Gamma_{\text{wk}}) \geq R_k^{\min}$ (private rates alone cover the demand), leaving power free to concentrate, so its serve-all frontier (3.11 bps/Hz) far exceeds the equal-split level of Eq. (45); in Case 2 the group load reaches the QoS capacity of Appendix A — serving all users under concentration requires each group's common budget to cover the withdrawn private rates, $(n_g - m_g) R_k^{\min} \leq R_{c,g}(\Gamma_{\text{min}}^g)$, the shortfall bound in its homogeneous form — which forces a near-equal split and flattens the frontier (serving $8 \rightarrow 10$ users costs only $1.80 \rightarrow 1.61$ bps/Hz). At matched QoS the trained agents attain $\approx 70\%$ of the frontier in Case 1 (both actors; the gap lies on the power-concentration axis) and $95\%/94\%$ (VQC/DNN) in Case 2 — consistent with the two operating points observed in Section V-B2.

TABLE XI: Pareto-frontier quantities per case (closed forms vs. the numerical per-state oracle, 500 states, environments of Tables IV–V).

Quantity	Case 1 ($K=5$)	Case 2 ($K=10$)
V_{rate}^* , Eq. (44) / oracle	10.963 / 10.963	11.723 / 11.723
Serve-all frontier $R^*(K)$ (bps/Hz)	3.11	1.61
Critical weight λ_{crit} , Eq. (46)	2.00	1.15
Training $\lambda_D = 1.5$ vs. λ_{crit}	below (single-user regime)	above (serve-all regime)
Frontier attainment at matched QoS (VQC / DNN)	$\approx 70\%$ / $\approx 70\%$	95% / 94%

APPENDIX C

FROZEN- λ ANALYSIS

Setup. The encoding scales $\lambda = (\lambda_y, \lambda_z) \in \mathbb{R}^{2n_q}$ of Section IV-C are trained jointly with the variational angles by SPSA. Across the entire Case-2 curriculum (four stages, 8000 episodes — the checkpoint chain behind Fig. 7) they exhibit no meaningful drift, a regime we refer to as *frozen* λ . This subsection quantifies the effect, identifies its cause, and bounds its cost.

Observation: empirical freeze. Fig. 9 tracks $\|\lambda_y\|_\infty$ and $\|\lambda_z\|_\infty$ over the full curriculum: both remain inside a 0.05-wide band for 8000 episodes. Between the first and the final checkpoint,

$$\rho_\lambda = \frac{\|\lambda_T - \lambda_0\|_2}{\|\lambda_0\|_2} = 0.14, \quad \|\lambda_T - \lambda_0\|_\infty = 0.109 \approx c_s = 0.1, \quad (47)$$

i.e., over the whole curriculum no single component of λ moves by more than approximately one SPSA perturbation step. The per-update diagnostics show why no progress accumulates: the mini-batch gradient magnitude is healthy ($\|\bar{\hat{\mathbf{g}}}_\lambda\| = 0.48$) while its batch mean is an order of magnitude smaller ($\|\bar{\mathbf{g}}_\lambda\| = 0.044$; ratio $r = 0.09$ on average, 0.14 at the 95th percentile).

Cause: gradient interference, not vanishing gradients. On the final checkpoint we recompute per-sample λ -gradients over 240 rollout samples. The per-sample magnitude is large, $\|\bar{\hat{\mathbf{g}}}_b\| = 10.4$ — far above the barren-plateau scale $O(2^{-n_q})$ [39] — yet the mean pairwise cosine between samples,

$$\mathbb{E}_{b \neq b'} [\cos(\hat{\mathbf{g}}_b, \hat{\mathbf{g}}_{b'})] = +0.009, \quad (48)$$

is essentially zero: individual samples carry strong but mutually uncorrelated directions, so their batch mean cancels and the first moment available to the optimizer stays near zero. Removing the advantage

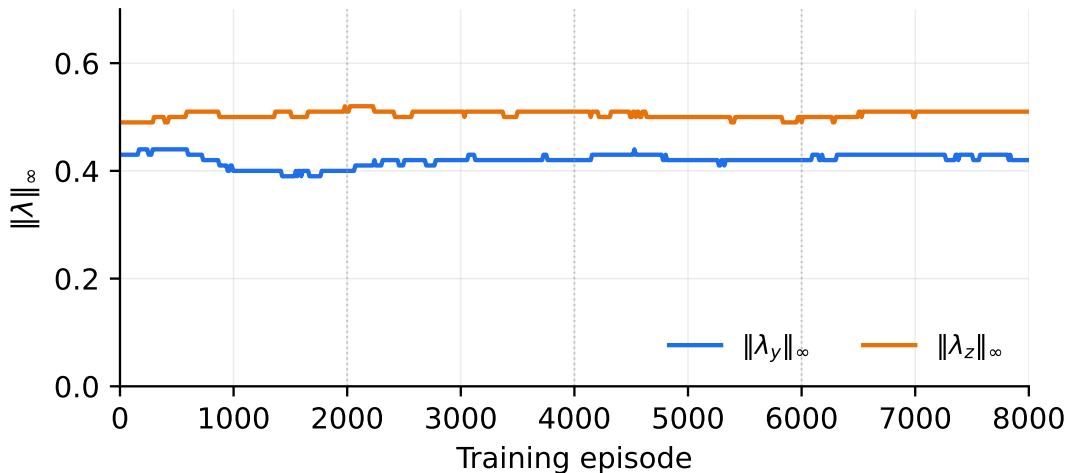


Fig. 9: Maximum absolute encoding scales $\|\lambda_y\|_\infty$, $\|\lambda_z\|_\infty$ across the full Case-2 curriculum (8000 episodes; dotted lines mark curriculum-stage boundaries). Both fluctuate marginally around their initial values and never drift.

sign raises the cosine to $+0.10$, indicating that a substantial part of the cancellation stems from sign flips of the advantage estimate rather than from the quantum landscape itself.

The frozen value is rescaling-optimal. Sweeping a global rescaling $\lambda \rightarrow c\lambda$, $c \in [0.1, 3]$, on the final checkpoint (60 frozen channel states $\times$ 5 seeds) finds the frozen value itself to be the best global factor in every seed ($\Delta(c^*) - \Delta(1) = 0.00$), with the per-state-optimal rescaling adding at most 0.01–0.04 against a reward scale of ~ 1.5 . Away from the frozen value the reward degrades sharply (at $c=2$: -1.07 to -3.09 across seeds): the encoding scales are far from irrelevant, but the surrounding network has co-adapted so that the frozen value sits at the optimum of the rescaling direction. Freezing therefore costs nothing measurable.

Encoder contribution beyond the classical bypass. Finally, we check that the encoding stage with frozen scales is not decorative. We attenuate the quantum encoder by a factor $c \in \{1, 0.75, 0.5, 0.25, 0\}$ (with $c=0$ leaving only the classical bypass through the post-processing head) and measure the policy’s state-discrimination signal — the mean pairwise KL divergence between the assignment distributions produced for distinct channel states. On the final checkpoint the signal drops from 7.06 (full encoder) to 5.30 (bypass only): an encoder-only share of 25%. The degradation is smooth in c (every attenuation step within 11% of the linear trend), so the contribution is a genuine continuous signal rather than a discontinuity artifact. Across the four checkpoints of the curriculum the absolute signal grows by $2.4\times$ while the encoder share declines from $\sim 50\%$ to $\sim 25\%$ — the classical bypass grows faster in late training, but the frozen-scale

TABLE XII: Frozen- λ verification on the Case-2 anchor (chain 8000 episodes; final checkpoint for the per-sample and rescaling probes).

Quantity	Value
Relative drift ρ_λ , Eq. (47)	0.14
$\ \lambda_T - \lambda_0\ _\infty$ vs. c_s	0.109 vs. 0.1
In-training $\ \hat{\mathbf{g}}_\lambda\ / \ \bar{\mathbf{g}}_\lambda\ $ / ratio r	0.48 / 0.044 / 0.09
Per-sample $\ \hat{\mathbf{g}}_b\ $ (240 samples)	10.4
Mean pairwise cosine, Eq. (48)	+0.009
Cosine with advantage sign removed	+0.10
Global-rescaling gain $\Delta(c^*) - \Delta(1)$ (5 seeds)	0.00
Per-state-optimal rescaling gain	≤ 0.04
Encoder-only share of state discrimination (final ckpt)	25% (smooth in attenuation)

encoder retains a substantial share throughout.

Taken together, the frozen- λ regime reflects an optimization property of this task — directionally incoherent per-sample gradients whose batch mean cancels, around a co-adapted optimum — rather than a degeneracy of the encoding stage itself; an honest, task-dependent contrast to the trainable-scale benefit reported in [15]. Table XII summarizes the verification quantities.

APPENDIX D

ARCHITECTURAL AND TRAINING ABLATIONS

A. Quantum vs. Classical Actor

Protocol. All comparisons swap only the high-level assignment actor and hold everything else fixed. The quantum pipeline — AE encoder, VQC, and SoftmaxPQC head — is replaced by a fully classical network of standard size: a plain MLP encoder ($d_s \rightarrow 128 \rightarrow 24$) followed by a policy MLP ($24 \rightarrow 256 \rightarrow 128 \rightarrow K(M+1)$ logits). The classical actor consumes the identical state features, emits the same latent interface $\mathbf{z}_t$ to the unchanged sub-actors, and is trained by the identical PPO recipe, seed, and episode budget. We report five axes: task performance and training budget, zero-shot generalization, model scalability, sample efficiency, and representation quality.

1) *Performance and Training Budget:* Across both scales the two actors reach sum rates within 3% of each other at two distinct operating points — the VQC leans toward QoS (+4.9/+1.9 pp at Cases 1/2), the DNN toward raw rate (Tables IV–V) — consistent with the prior that parametrized quantum policies match, rather than exceed, classical ones on natural control tasks [15]. The training budgets differ by

TABLE XIII: Zero-shot distribution shift (Case 2): both actors evaluated without retraining under test-time environment overrides. Base = the operating point of Table V ($R_{\text{LoS}}=0.5$, $P_S=50$ dBm, $\kappa=0.05$, $\sigma^2=10$ dB², speed 1.5 m/s, balanced spawn); greedy inference, 50 episodes, seed 42. Per-episode sum-rate standard deviations lie within 0.0014–0.0027 (VQC) and 0.0027–0.0055 (DNN) across all shifts; the VQC–DNN gaps exceed these fluctuations.

Shift	HQC-HAC (VQC)		DNN	
	R_{tot}	QoS	R_{tot}	QoS
base	1.5307	97.3%	1.5182	95.4%
$P_S = 40$ dBm	1.2947	74.5%	1.2654	70.5%
$P_S = 45$ dBm	1.4462	87.1%	1.4268	83.4%
$P_S = 60$ dBm	1.5818	99.6%	1.5777	99.2%
speed 3 m/s	1.5244	97.1%	1.5120	95.0%
speed 6 m/s	1.4928	95.3%	1.4840	93.2%
natural spawn	1.5312	97.2%	1.5122	95.3%
$\kappa = 0.10$	1.5291	97.2%	1.5170	95.4%
$\kappa = 0.20$	1.5263	97.0%	1.5150	95.3%
$\sigma^2 = 13$	1.5270	96.8%	1.5150	94.8%
$\sigma^2 = 16$	1.5241	96.3%	1.5119	94.3%

the cost of classical simulation: 23–37 s per episode for the VQC actor against 0.7–1.1 s for the DNN (median over the final 100 episodes, identical machine), a $\sim 30\times$ overhead incurred by $\approx 38,000$ statevector evaluations of a 2^{12} -amplitude register per episode rather than by the algorithm itself; on quantum hardware the corresponding cost unit is measurement shots. Simulated inference costs ≈ 0.17 versus ≈ 0.008 s per decision step.

2) *Generalization*: Table XIII evaluates both trained Case-2 actors *zero-shot* — no retraining — under test-time environment shifts. Both degrade gracefully on every axis except transmit power, which switches the regime from interference- to power-limited (Appendix B); the VQC’s QoS margin is preserved under every shift (+1.7 to +2.1 pp) and widens under power stress (+3.7/+4.0 pp at $P_S=45/40$ dBm), while unbalanced (natural) user spawning is virtually free for both actors. The element count N admits no zero-shot transfer, as the phase sub-actor’s output dimension is tied to N ; N -robustness therefore requires per- N retraining (Table VII). All entries are single-seed and we do not read the margins as systematic.

3) *Model Scalability*: The assignment-specific parameter budget separates the actors by orders of magnitude at matched task performance. The quantum decision core — variational angles, encoding scales, and encoding-normalization affine —

$$|\theta_{\text{VQC}}| = \underbrace{2n_q L}_{\theta^y, \theta^z} + \underbrace{2n_q}_{\lambda_y, \lambda_z} + \underbrace{4n_q}_{\gamma, \mathbf{b}_{\text{LN}}} = 2n_q(L+3) = \mathcal{O}(n_q L), \quad (49)$$

counts 120/144 parameters at Cases 1/2 against 40,586/43,166 for the classical policy MLP of matched role — a $338\times/300\times$ reduction that grows only linearly in the register size, whereas a classical block of matched input–output dimensions scales as $\mathcal{O}(d_h(2n_q+N_Q))$ in its hidden width d_h . At the full inference-time actor level (AE decoder discarded at deployment) the counts are 1,063/2,761 against 45,986/52,022 ($43\times/19\times$). The stated ratios are specific to the chosen classical width. A first iso-performance point — a 2,230-parameter policy MLP trained under the identical recipe at the $R_{\text{LoS}}=0.2$ stage — already matches both the 40K classical baseline and the VQC (1.516/97.4% against 1.509/96.5% and 1.512/96.8%, single seed), indicating that the standard-width baseline is over-parameterized for this task and that the matched-performance ratio of assignment-decision parameters is nearer $15\times$ (2,230 versus 144); the remaining width grid is left to future work.

4) *Sample Efficiency*: Both actors train under the identical PPO recipe and episode budget, so differences in learning speed are attributable to the actor alone. At the two hardest curriculum transitions (resuming into $R_{\text{LoS}}=0.4$, and into 0.5 with full spawn unlock), both actors recover to within 90% of their final QoS at the same speed (~ 25 episodes), but the transient dip is consistently shallower for the VQC: minimum sampled QoS 87%/91% at the two transitions, against 83%/84% for the DNN. The curriculum-robustness advantage therefore lies in the depth, not the duration, of the transient (single seed).

5) *Representation Quality*: A distillation probe quantifies how much assignment-relevant information the latent $\mathbf{z}_t$ carries, by supervised recovery of the oracle assignment from frozen latents. For the quantum pipeline, a lightweight classical probe recovers 51% of the oracle-assignment reward gap from $\mathbf{z}_t$, against 27% from the raw state and none by the live policy during standard training — the latent concentrates, rather than merely compresses, the assignment-relevant information. The classical-actor counterpart of this probe is pending: —.

B. Autoencoder Latent Compression

The AE compresses the high-dimensional state into a latent $\mathbf{z}_t \in \mathbb{R}^{2n_q}$ that (a) is necessary to fit the input into the n_q -qubit register, and (b) concentrates, rather than merely preserves, the assignment-relevant information. We assess (i) the qubit cost of encoding without compression against the fixed $n_q=12$ with AE; and (ii) the information loss of the fixed-width latent as the problem scales, governed by the

TABLE XIV: Encoding cost and compression per case. The VQC input dimension $d_{\text{aff}} = K(M+2)+2M$ sets the register a raw, AE-free angle encoding would need, $n_q^{\text{raw}} = \lceil d_{\text{aff}}/2 \rceil$ (two rotations per qubit); the AE fixes it at $n_q=12$. L_{AE} is the converged reconstruction floor (Fig. 10); Case 3 is untrained.

(K, M, N)	d_{aff}	n_q^{raw}	n_q^{AE}	L_{AE}
(5, 1, 24)	17	9	12	2×10^{-4}
(10, 2, 24)	44	22	12	5×10^{-3}
(15, 3, 24)	81	41	12	—

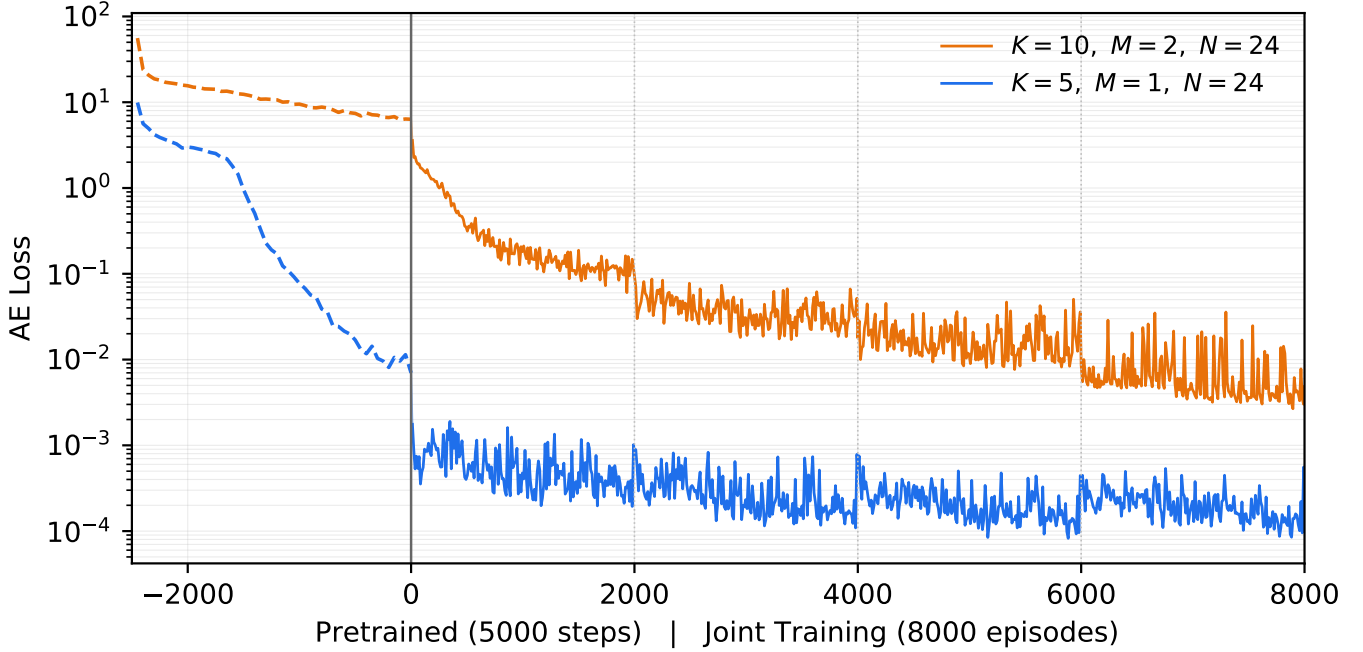


Fig. 10: AE reconstruction loss L_{AE} : pre-training (dashed, 5000 steps on random states, left of the solid vertical line) hands off to joint training over the full R_{LoS} curriculum (solid, $0.2 \rightarrow 0.5$; dotted lines mark ramp transitions; log scale). Pre-training on random inputs floors at $L_{AE} \approx 7 \times 10^{-3}$ (Case 1) and ≈ 6.3 (Case 2) — a heuristic starting point; joint training on the structured rollout states then drives L_{AE} to $\sim 2 \times 10^{-4}$ under expansion (Case 1, $d_{\text{aff}}=17 < 24$, near machine-lossless) and to a floor $\sim 5 \times 10^{-3}$ under compression (Case 2, $d_{\text{aff}}=44 > 24$) — the floor is reached, not traded away for task loss.

compression ratio $d_{\text{aff}}/2n_q$ with $d_{\text{aff}} = K(M+2)+2M$ growing in K and M (N is filtered out by the affinity construction).

C. VQC Architectural Choices

Five architectural knobs define the quantum actor; each is varied one at a time against the reference configuration ($n_q=12$, $L=2/3$, nearest-neighbor CZ chain with M cross-block bridges, $R_y R_z$ angle encoding with data re-uploading, R1 readout).

TABLE XV: Readout-design ladder (Case 1/Case 2), identical hyperparameters, greedy inference. The observable count grows $\text{single-}Z \rightarrow Z+\text{NN-}ZZ \rightarrow Z+\text{full-}ZZ \rightarrow \mathbf{R1}$; only the readout differs. R1 is the structured set of Eq. (21).

Readout	$ \hat{o} $ (C1/C2)	R_{tot} (C1/C2)	QoS (C1/C2)
single- Z	12/12	1.7890/1.5025	99.9%/97.1%
$Z+\text{NN-}ZZ$	23/23	1.7715/1.5072	99.9%/96.5%
$Z+\text{full-}ZZ$	33/41	1.7282/1.5012	99.7%/95.5%
R1 (structured)	34/42	1.7471/1.5017	99.7%/96.8%

1) *Circuit Depth*: The number of variational layers L widens the accessible frequency spectrum of the re-uploading circuit [40], so performance is expected to improve with L up to saturation. The depth sweep is pending: —.

2) *Entanglement Pattern*: Candidate patterns of increasing connectivity — none, linear chain, ring, full — are compared against the reference chain augmented with M cross-block CZ bridges. The bridges are the problem-specific element: they couple the user- and IRS-qubit registers and are hypothesized to carry the user–IRS interaction that the assignment must model, whereas denser generic patterns add entanglement without this alignment. The toggle runs are pending: —.

3) *Qubit Count and Latent Dimension*: The register size n_q fixes the latent width $2n_q$ and hence the compression ratio $d_{\text{aff}}/2n_q$; it trades representation capacity against the 2^{n_q} simulation cost. A preliminary Case-1 comparison found $n_q=8$ to underperform the unified $n_q=12$ configuration by a wide QoS margin; a controlled sweep is pending: —.

4) *Feature Map*: The reference encoding loads each latent pair through $R_y R_z$ rotations with a tanh squash and re-applies the encoding block at every layer (data re-uploading [38]). Alternative feature maps (amplitude encoding, IQP-style maps) change the accessible function class and the loading cost. A comparison is pending: —.

5) *Readout Design*: The readout ladder sweeps Pauli observable sets of increasing richness at otherwise identical hyperparameters; Table XV reports both cases.

D. Entropy Regularization and Annealing

Hypothesis 5. *The annealed entropy bonus of (24) is necessary to explore the combinatorial assignment space: a fixed-low β converges prematurely, a fixed-high β fails to exploit, and the $\beta_0 \rightarrow \beta_{\min}$ schedule dominates both.* The entropy gradient $\nabla H(\pi_j) = -\nabla \sum \pi_j (1 + \log \pi_j)$ pushes each actor toward uniform

exploration with strength β , mirroring the “trainable greediness” role of the inverse temperature in softmax-PQC policies [15]. **Setup.** Compare fixed-low, fixed-high, and annealed β (reference); report exploration proxies (assignment entropy, IRS-usage) alongside sum-rate. **Observations.**

- The training logs bear out the hypothesis qualitatively: a fixed-high β holds the assignment distribution near-uniform so the policy never commits (the “entropy-dominated” regime), whereas a fixed-low β collapses early onto a sub-optimal assignment; the $\beta_0 \rightarrow \beta_{\min}$ schedule — which explores first and exploits later — is used throughout and avoids both failure modes. A controlled fixed-versus-annealed sweep at matched seeds is left to future work.

E. Power-Fairness Blending

Hypothesis 6. *The power-fairness coefficient $\alpha_{\text{pf}} \in [0, 1]$, which blends the executed split/common/private power toward the QoS-optimal equal allocation while leaving the sampled action and its log-probability unchanged, is a monotone Rate–QoS control knob whose QoS-optimal operating point is case-specific rather than universal.* Concretely, the Power MLP’s categorical split/common/private probabilities (p_s, p_c, p_p) still parameterize the policy gradient unaltered; only the powers used to *execute* the environment step are blended, $p^{\text{eff}} = (1 - \alpha_{\text{pf}})p + \alpha_{\text{pf}}p^*$, where p^* targets a fixed private-vs-common split and a uniform distribution within each axis. $\alpha_{\text{pf}}=0$ recovers the raw learned allocation and $\alpha_{\text{pf}}=1$ forces the QoS-favoring equal split, so α_{pf} trades sum-rate concentration for QoS coverage without touching the gradient path. **Setup.** Sweep α_{pf} with every other hyperparameter fixed (identical seed, recipe, and curriculum): $\alpha_{\text{pf}} \in \{0.3, 0.4, 0.5\}$ on Case 1 and $\alpha_{\text{pf}} \in \{0.5, 0.7, 0.9\}$ on Case 2; report greedy-inference R_{tot} and QoS for each point. **Observations.**

- *Power-fairness sweep* (Table XVI). Runs pending at the time of writing; only the endpoints already used as the Case-1 ($\alpha_{\text{pf}}=0.5$, Table IV) and Case-2 ($\alpha_{\text{pf}}=0.9$, Table V) main-result operating points are available. We expect QoS to rise monotonically in α_{pf} at a mild sum-rate cost, consistent with α_{pf} pushing power away from the rate-maximizing concentration and toward equal coverage; the QoS-optimal α_{pf} is expected to sit higher for Case 2 than Case 1, since a larger K leaves proportionally more users exposed to the worst-case shortfall that equal power directly relieves. Because the two cases were locked at different endpoints of the sweep (0.5 vs. 0.9) rather than a shared grid, the frontier is case-specific and should not be read as a single universal knob value.

TABLE XVI: Power-fairness sweep (Case 1 / Case 2), identical hyperparameters and curriculum except α_{pf} , greedy inference.

α_{pf}	R_{tot}	QoS
C1, 0.3	—	—
C1, 0.4	—	—
C1, 0.5	—	—
C2, 0.5	—	—
C2, 0.7	—	—
C2, 0.9	—	—

F. Warm-Up Strategy Isolation

Hypothesis 7. *Warming the assignment and/or phase sub-policies on oracle targets before the PPO objective is switched on suppresses the early reward “yank” seen from a cold start; isolating the two warm-up paths shows the assignment (routing) warm-up carries most of the benefit, while the phase warm-up saturates once the power blend of Appendix D-E is set.* **Setup.** At fixed α_{pf} on Case 2, compare four warm-up configurations: (i) vanilla (no warm-up); (ii) assignment-only, warming the quantum actor’s Q-head toward the oracle blocked-user routing target; (iii) phase-only (`--phase-warmup`), warming the Phase MLP toward the closed-form phase oracle; (iv) dual (`--oracle-warmup`), warming both assignment and phase jointly. Metrics: greedy-inference R_{tot} and QoS, plus a convergence-speed proxy (episodes to reach 90% of final QoS). **Observations.**

- *Warm-up isolation* (Table XVII). Runs pending. (i), (iii), and (iv) are launch-ready with existing flags; (ii) currently has no dedicated flag — `--oracle-warmup` warms assignment and phase jointly and cannot isolate the Q-head path alone, so an assignment-only warm-up requires adding a flag that runs the assignment branch of the existing oracle-warmup routine without the phase branch before this row can be filled. We expect the ordering (ii) $\approx$ (iv) $>$ (iii) $\approx$ (i): the routing bootstrap is expected to dominate the benefit (consistent with the oracle-intervention finding that substituting the assignment yields the largest single-actor gain), while the phase warm-up alone saturates and contributes comparatively little once the power blend is in place.

TABLE XVII: Warm-up strategy isolation (Case 2), identical hyperparameters and curriculum except the warm-up configuration, greedy inference. The dual row is the canonical Case-2 operating point ($R_{\text{LoS}}=0.2$ stage of the Table V chain); the remaining configurations are pending.

Warm-up	R_{tot}	QoS	Convergence
Vanilla (none)	—	—	—
Assignment-only	—	—	—
Phase-only (<code>--phase-warmup</code>)	—	—	—
Dual (<code>--oracle-warmup</code>)	1.512	96.8%	—